

A counterexample to integer-degree harmonic dimension comparison

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Abstract

For some even $n \geq 8$ and integer $k \geq 2$, we construct a complete smooth metric on $\mathbb{R}^n$ with nonnegative Ricci curvature whose space of real harmonic functions of pointwise polynomial growth at most k has dimension larger than the Euclidean harmonic-polynomial dimension. The metric is Euclidean near the origin, has asymptotic volume ratio strictly between zero and one, and has nonunique tangent cones at infinity. This answers the integer-degree form of Yau's dimension comparison question in the negative.

1 Introduction

For a complete connected smooth Riemannian manifold (M^n, g) without boundary, let $\mathcal{H}_d(M, g)$ be the real vector space of harmonic functions satisfying

$$|u(x)| \leq C_u(1 + d_g(o, x))^d \quad (x \in M)$$

for some finite constant C_u , where $d \geq 0$ and $o \in M$ is a base point. Changing o does not change the space. Write $h_d(M, g) = \dim_{\mathbb{R}} \mathcal{H}_d(M, g)$. On Euclidean space, an entire harmonic function with integer growth at most k is a harmonic polynomial of degree at most k . Consequently

$$h_k(\mathbb{R}^n, g_E) = \binom{n+k-1}{k} + \binom{n+k-2}{k-1} \quad (k \geq 1). \quad (1.1)$$

The integer-degree comparison problem asks whether $\text{Ric}_g \geq 0$ forces $h_k(M, g) \leq h_k(\mathbb{R}^n, g_E)$ for every integer $k \geq 1$. The condition defining $\mathcal{H}_k$ is pointwise at every distance. Bounds only along a sequence of spheres do not establish it.

Theorem 1.1. *There are an even integer $n \geq 8$, an integer $k \geq 2$, and a complete smooth metric g on $\mathbb{R}^n$ such that $\text{Ric}_g \geq 0$ and*

$$h_k(\mathbb{R}^n, g) > h_k(\mathbb{R}^n, g_E).$$

The metric is Euclidean near the origin, and its asymptotic volume ratio satisfies

$$0 < \lim_{R \rightarrow \infty} \frac{\text{vol}_g B_g(0, R)}{\omega_n R^n} < 1,$$

where ω_n is the volume of the Euclidean unit ball.

The exact finite spectral selection in Appendix A permits the choices $n = 16$ and $k = 50000$.

Theorem 1.1 answers the integer-degree form of Yau's dimension comparison question in the negative. Yau's questions on polynomial-growth harmonic functions distinguish finite dimensionality from the sharper Euclidean comparison [17, 13]. Li and Tam proved the sharp linear-growth bound $h_1(M, g) \leq n + 1$ under nonnegative Ricci curvature [13]. Colding and Minicozzi proved finite dimensionality at every fixed degree and later obtained bounds of the optimal order d^{n-1} as d grows [3, 4]. Those estimates leave open the exact Euclidean count at a particular integer degree.

Donnelly constructed counterexamples to the analogous comparison at a real degree strictly between one and two in dimensions at least five [6, 2]. That fact alone does not decide the integer endpoint two: the Euclidean space of quadratic harmonic polynomials is larger than its space at a degree between one and two. Cai and Lai make this distinction explicit and establish sharp comparison with the additional hypothesis of local conformal flatness [2]. The theorem above addresses the integer question under the Ricci hypothesis alone.

Cone-spectral results explain another boundary of the problem. For complete manifolds with nonnegative Ricci curvature, maximal volume growth, and a unique tangent cone at infinity, Huang's Theorem 1.6 relates the harmonic dimension to the spectrum of the cone link [8]. Our metric instead has round and nonround subsequential cone limits. Nonunique cones with positive Ricci curvature and Euclidean volume growth already occur in Perelman's construction [14]; Colding and Naber developed a method for slowly varying link families with a common volume form [5]. We use the same geometric possibility of a moving link, while proving the curvature estimates for our particular deformations. Xu's nonunique-cone three-circles theorem requires a comparison exponent outside the union of cone-degree spectra [16, Theorem 3.4]. A separate geometric conclusion of our construction is that this union contains a neighborhood of the selected integer k , so the theorem cannot be applied at k or at sufficiently nearby exponents.

Why a moving link can change the count

Put $n = m + 1$. On the metric cone $dr^2 + r^2h$ over an m -dimensional link, an angular eigenfunction of the nonnegative Laplacian with eigenvalue λ has a growing homogeneous harmonic mode $r^d\phi$, where

$$d(d + m - 1) = \lambda, \quad d \geq 0.$$

For one fixed link, each angular direction therefore carries one fixed growth rate. Our construction lets a harmonic function use several rates in succession. The challenge is to arrange that succession exactly: an arbitrarily small component in an unwanted faster direction can dominate after a sufficiently long interval.

The links are volume-normalized Berger metrics on odd spheres S^m , $m = 2s - 1$. Their distinguished Hopf direction is allowed to change. Differentiation along this circle direction is the Hopf generator; on complexified harmonics its eigenvalues are i times the integer Hopf charges. Every angular Laplacian nevertheless preserves the same finite-dimensional space V_l of round spherical harmonics of degree l ; write $N_l = \dim_{\mathbb{R}} V_l$. At a fixed nonround link, write its growing rates on V_l in increasing order as $d_{l,1}, \dots, d_{l,N_l}$. The spectral calculation supplies a finite $L \geq 2$ and integers $0 \leq M_l \leq N_l$ for $2 \leq l \leq L$ such that

$$\sum_{l=2}^L M_l > h_k(\mathbb{R}^{m+1}, g_E), \quad \frac{1}{M_l} \sum_{i=1}^{M_l} d_{l,i} < k \quad (M_l > 0). \quad (1.2)$$

Thus the count will be large enough if each selected harmonic function can spend asymptotically

equal amounts of logarithmic radius at all the selected rates in its block. The inequality for the mean is strict; individual rates may exceed k .

The source of (1.2) is the distribution of Hopf charges inside V_l . The squared charges divided by l^2 tend to a beta distribution. The mean of the lowest fraction of rates is described by an integral of its decreasing quantile. In a sufficiently large odd link dimension, a weighted quantile inequality shows that selecting directions by their mean produces a strict surplus over the Euclidean count. This fixes the link parameters, the integer k , and a finite degree range before the radial construction begins. An exact-integer computation gives a separate finite selection at specific parameters; its role is to verify that finite spectral input, rather than the subsequent geometric or transmission proof.

From angular control to entire harmonic functions

Short deformations near a round link provide the exchanges of directions. For every fixed finite range $2 \leq l \leq L$, the trace-free squares of the Hopf generators, using all orthogonal complex structures, generate

$$\bigoplus_{l=2}^L \mathfrak{sl}(V_l).$$

The real form and the direct sum matter: one must choose the same finite sequence of geometric pulses while independently arranging a prescribed signed permutation in each V_l . A finite product of exponentials realizes these permutations at a parameter value where the word map has invertible differential. The transmission argument below shows that this algebraic freedom survives the harmonic equation.

Write the metric as $g = dr^2 + r^2\gamma(t)$ with $t = \log r$. A harmonic function in V_l is governed by a second-order matrix equation. Smoothness at the Euclidean center selects a distinguished regular solution space. Its logarithmic derivative P_l satisfies a matrix Riccati equation; positive barriers keep P_l bounded and symmetric. A long round rest makes P_l approach a scalar matrix, independently of all earlier admissible pulses. During the next control interval, which is short relative to the following dwell, the determinant-normalized value transfer converges in C^1 to the finite algebraic word as the period index tends to infinity.

The pulse-end approximation is only an intermediate fact: the outgoing radial derivative can still inject a small component into a faster direction during the following long leg, on which the angular eigenvectors remain fixed. We factor out the positive diagonal transfer $D_{l,j}$ obtained by entering that entire leg with the round growing derivative. The remaining multiplier of the derivative error is bounded independently of the leg's length and anisotropy (Lemma 6.4). Thus the full-period map, with this diagonal factor removed and its determinant normalized, is C^1 -close to the ideal word, uniformly over all preceding controls. Its local inverse tunes every complete period exactly to $s_{l,j}D_{l,j}\Pi_l$, where Π_l cycles the selected axes and $s_{l,j} > 0$ is a scalar.

The geometry and the growth argument use different features of the same schedule. In period j , the control intervals have length comparable to j^6 and the nonround dwell has length j^7 , while the accumulated logarithmic radius is comparable to j^8 . Angular motion is consequently slow enough for a small concave radial tail to preserve nonnegative Ricci curvature. The diagonal leg contributes $\log(D_{l,j})_{ii} = j^7(d_{l,i} + o(1))$, while $\log s_{l,j} = O(j^6)$. Exact cycling and the asymptotic equality of finitely many consecutive dwell lengths yield the means in (1.2). Bounded Riccati matrices control the intervals between period endpoints. Since a whole period is negligible compared with the accumulated logarithmic radius, the same exponent bound holds at every radius. The strict

inequality below k then gives the pointwise growth condition, and invertibility of the regular fundamental matrices gives the claimed number of independent smooth harmonic functions.

All spaces and functions are real unless explicitly complexified. Angular Laplacians are nonnegative. Matrices act on column vectors, and positivity of a matrix means positivity as a symmetric operator. Constants may depend on the fixed finite degree range and fixed compact family of controls; they are uniform in the period index and in the preceding admissible controls.

Sections 2 and 3 establish the Berger formulas and the finite spectral surplus. Section 4 proves simultaneous real control, and Section 5 realizes every admitted control history by a complete metric with nonnegative Ricci curvature. Section 6 tunes the full-period transmissions exactly; Section 7 proves the every-point growth estimate and the cone conclusions. Appendix A records the independent finite arithmetic selection.

2 Volume-normalized Berger links

We need angular metrics that retain fixed finite-dimensional harmonic spaces while varying their spectra. This section derives their curvature and angular operators; the next section selects the directions that will be cycled.

Fix an odd integer $m = 2s - 1$, with $s \geq 4$, and let g_0 be the unit round metric on $S^m \subset \mathbb{R}^{2s}$. An orthogonal complex structure is a real linear map J satisfying $J^2 = -I$ and $J^\top J = I$. It determines the unit round Killing field and its dual one-form

$$\xi_J(x) = Jx, \quad \eta_J = g_0(\xi_J, \cdot).$$

Write

$$P_J = \xi_J \otimes \eta_J, \quad D_J u = du(\xi_J).$$

Thus P_J is the projection onto the Hopf direction. For $a, q > 0$, define

$$\tilde{g}_{q,J} = g_0 + (q - 1)\eta_J^2, \quad G(a, q, J) = a^2 q^{-1/m} \tilde{g}_{q,J}.$$

The factor $q^{-1/m}$ makes the volume independent of q . All orthogonal complex structures will be allowed; we do not restrict them to one component of their parameter space.

This volume normalization and the corresponding Laplacian decomposition are classical; see Tanno [15, Sections 2–4]. His parameter t corresponds to $q = t^m$ when $a = 1$. Classical curvature calculations for these spheres appear in Bourguignon–Karcher [1, Section 6.6]. We record the formulas and their direct proofs in our parameters below.

For $l \geq 0$, let V_l be the real vector space of restrictions to S^m of homogeneous harmonic polynomials of degree l on $\mathbb{R}^{m+1}$. Equip V_l with the round L^2 inner product and put

$$N_l = \dim_{\mathbb{R}} V_l = \binom{m+l}{m} - \binom{m+l-2}{m}, \quad B_l = l(l+m-1),$$

where the second binomial coefficient is zero for $l < 2$. For completeness, on homogeneous polynomials give distinct monomials orthogonal inner products with $\langle x^\alpha, x^\alpha \rangle = \alpha!$. The adjoint of the Euclidean polynomial Laplacian from degree l to degree $l-2$ is multiplication by $|x|^2$, which is injective. The Laplacian is therefore surjective, giving the dimension formula. Restriction to the sphere is injective on homogeneous polynomials. Euclidean polar coordinates give

$$\Delta_{g_0}^+|_{V_l} = B_l I, \quad \Delta_h^+ := -\operatorname{div}_h \nabla_h.$$

Rotations preserve V_l . In particular, D_J preserves V_l and is skew-adjoint for its round inner product.

Lemma 2.1. *The volume form of $G(a, q, J)$ is $a^m d\text{vol}_{g_0}$. Its Ricci endomorphism has eigenvalues*

$$a^{-2}q^{1/m}((m-1) - 2(q-1)) \quad \text{horizontally,} \quad a^{-2}q^{1/m}(m-1)q \quad \text{vertically.}$$

Consequently,

$$\text{Ric}_{G(a,q,J)} > (m-1)G(a, q, J) \iff a^2 < q^{1/m} \min \left\{ 1 - \frac{2(q-1)}{m-1}, q \right\}. \quad (2.1)$$

Every V_l is preserved by the positive Laplacian of $G(a, q, J)$, whose restriction is

$$A_l(a, q, J) = a^{-2}q^{1/m} [B_l I + (q^{-1} - 1)(-D_J^2|_{V_l})]. \quad (2.2)$$

Proof. We suppress J from the notation. Relative to g_0 , the metric $\tilde{g}_q$ has eigenvalues 1 on the $(m-1)$ -dimensional horizontal space and q on the Hopf direction. Its volume form is $\sqrt{q} d\text{vol}_{g_0}$. Multiplication of the metric by $a^2 q^{-1/m}$ multiplies this volume form by $a^m q^{-1/2}$, proving the volume assertion.

Here is a direct curvature calculation. Let ∇ be the round connection and set $\phi X = \nabla_X \xi$. Tangent projection of the ambient derivative gives

$$\phi X = JX + \eta(X)x, \quad \phi^* = -\phi, \quad \phi\xi = 0, \quad \phi^2 = -I + \xi \otimes \eta,$$

and differentiation gives

$$(\nabla_X \eta)(Y) = g_0(\phi X, Y), \quad (\nabla_X \phi)Y = \eta(Y)X - g_0(X, Y)\xi.$$

Put $\delta = q - 1$, which is constant on the sphere. If $C = \tilde{\nabla} - \nabla$ is the difference between the connections of $\tilde{g}_q$ and g_0 , the Koszul formula yields

$$\begin{aligned} 2\tilde{g}_q(C(X, Y), Z) &= (\nabla_X \tilde{g}_q)(Y, Z) + (\nabla_Y \tilde{g}_q)(X, Z) - (\nabla_Z \tilde{g}_q)(X, Y) \\ &= 2\delta [\eta(X)g_0(\phi Y, Z) + \eta(Y)g_0(\phi X, Z)]. \end{aligned}$$

The vectors $\phi X, \phi Y$ are horizontal, so

$$C(X, Y) = \delta(\eta(X)\phi Y + \eta(Y)\phi X).$$

In particular, $\text{tr}[X \mapsto C(X, Z)] = 0$. Tracing the connection change formula for curvature therefore gives

$$(\text{Ric}_{\tilde{g}_q} - \text{Ric}_{g_0})(Y, Z) = \text{tr}[X \mapsto (\nabla_X C)(Y, Z)] - \text{tr}[X \mapsto C(Y, C(X, Z))].$$

These are endomorphism traces and can be computed in a round orthonormal frame. Using the identities for ϕ , the four terms obtained by differentiating C contribute, in order,

$$-\delta(g_0 - \eta^2), \quad \delta(m-1)\eta^2, \quad -\delta(g_0 - \eta^2), \quad \delta(m-1)\eta^2.$$

Thus the first trace is $\delta(-2g_0 + 2m\eta^2)(Y, Z)$. For the second trace, one has

$$C(Y, C(X, Z)) = \delta^2 \eta(Y) [-\eta(X)Z - \eta(Z)X + 2\eta(X)\eta(Z)\xi],$$

whose trace in X is $-(m-1)\delta^2 \eta(Y)\eta(Z)$. Since $\text{Ric}_{g_0} = (m-1)g_0$, this proves the tensor identity

$$\text{Ric}_{\tilde{g}_q} = ((m-1) - 2(q-1))(g_0 - \eta^2) + (m-1)q^2\eta^2.$$

The squared length of ξ in $\tilde{g}_q$ is q . Hence the vertical eigenvalue of its Ricci endomorphism is $(m-1)q$, while the horizontal eigenvalue is $(m-1) - 2(q-1)$. Constant metric scaling leaves the covariant Ricci tensor unchanged and divides its endomorphism eigenvalues by the scale. This proves the stated eigenvalues and (2.1).

Finally, the inverse metric is

$$G(a, q, J)^{-1} = a^{-2} q^{1/m} [g_0^{-1} + (q^{-1} - 1)\xi \otimes \xi].$$

Because its volume form is a spatially constant multiple of round volume, its divergence on vector fields is the round divergence. The field ξ has zero round divergence, so $\text{div}_{g_0}((DJu)\xi) = D_J^2 u$. It follows that

$$\Delta_{G(a,q,J)}^+ u = a^{-2} q^{1/m} [\Delta_{g_0}^+ u + (q^{-1} - 1)(-D_J^2 u)].$$

Restricting this identity to V_l gives (2.2). $\square$

3 A strict spectral surplus

Fix the standard orthogonal complex structure J_0 on $\mathbb{R}^{2s}$. For $a, q > 0$, let $\lambda_{l,1}(a, q) \leq \dots \leq \lambda_{l,N_l}(a, q)$ be the eigenvalues, counted with their real multiplicities, of the positive angular operator in (2.2) for $G(a, q, J_0)$. Define the corresponding nonnegative exponents by

$$d_{l,i}(a, q)(d_{l,i}(a, q) + m - 1) = \lambda_{l,i}(a, q), \quad d_{l,i}(a, q) \geq 0. \quad (3.1)$$

We will find more than the Euclidean number of directions whose exponents have mean strictly below a common integer. The subsequent construction will realize these means by cycling the selected directions.

The Hopf-charge decomposition underlying this calculation is described by Tanno [15, Lemmas 3.1 and 4.1]; we prove the exact multiplicities and the distribution needed for the count.

Lemma 3.1 (Hopf-weight distribution). *For $m = 2s - 1$ and $0 \leq b \leq l$, the eigenvalue $(2b - l)^2$ of $-D_{J_0}^2$ on V_l has the following contribution to its multiplicity:*

$$\begin{aligned} h_{l,b} &= \binom{s+b-1}{s-1} \binom{s+l-b-1}{s-1} - \binom{s+b-2}{s-1} \binom{s+l-b-2}{s-1} \\ &= \frac{l+s-1}{s-1} \binom{b+s-2}{s-2} \binom{l-b+s-2}{s-2}. \end{aligned} \quad (3.2)$$

Terms with a missing bidegree in the first line are zero. With probabilities $h_{l,b}/N_l$, the quantities $(2b - l)^2/l^2$ converge in distribution, as $l \rightarrow \infty$, to

$$Y_m \sim \text{Beta}\left(\frac{1}{2}, \frac{m-1}{2}\right).$$

Proof. Complexify V_l and decompose homogeneous polynomials into bidegrees $(b, l-b)$ in $z_1, \dots, z_s$ and their conjugates. The operator D_{J_0} acts on this bidegree by $i(2b-l)$. The contraction $\sum_j \partial_{z_j} \partial_{\bar{z}_j}$ maps onto the polynomials of bidegree $(b-1, l-b-1)$: in the positive monomial inner product in which $z^\alpha \bar{z}^\beta$ has squared norm $\alpha! \beta!$, its adjoint is multiplication by $\sum_j z_j \bar{z}_j$, which is injective. Taking the difference of dimensions gives the first line of (3.2); cancellation gives the second. The contributions indexed by b and $l-b$ belong to the same eigenvalue of $-D_{J_0}^2$ and are added; the middle contribution is counted once. Since this operator is real symmetric, the complex

dimension of each complexified eigenspace equals the real dimension of the corresponding original real eigenspace. There is no additional factor of two.

Summing the second line by the binomial convolution identity gives

$$N_l = \frac{l+s-1}{s-1} \binom{l+2s-3}{2s-3} = \binom{m+l}{m} - \binom{m+l-2}{m} \sim \frac{2}{(m-1)!} l^{m-1}. \quad (3.3)$$

More precisely, $h_{l,b}/N_l$ is a beta-binomial probability. If X has density proportional to $[x(1-x)]^{s-2}$ on $(0,1)$ and, conditionally on X , K_l is binomial with parameters l, X , then integration of the binomial probability against this density gives $\mathbb{P}(K_l = b) = h_{l,b}/N_l$. Since

$$\mathbb{E} \left[\left(\frac{K_l}{l} - X \right)^2 \right] = \frac{\mathbb{E}[X(1-X)]}{l} \leq \frac{1}{4l},$$

we have $K_l/l \rightarrow X$ in probability in this representation. The change of variable $y = (2x-1)^2$ gives density proportional to $y^{-1/2}(1-y)^{s-2}$, proving the assertion. $\square$

For $0 < v < 1$, define the upper-tail quantile $y_m(v) \in (0,1)$ by

$$\mathbb{P}(Y_m > y_m(v)) = v, \quad \mathcal{J}_m := \int_0^1 y_m(v) \log(1/v) dv.$$

The threshold $2/(m-1)$ in the next lemma comes from the horizontal factor in the link curvature condition (2.1).

Lemma 3.2 (A weighted quantile inequality). *For all sufficiently large odd m ,*

$$(m-1)\mathcal{J}_m > 2. \quad (3.4)$$

Proof. Put $r = m-1$. The beta density, equivalently polar coordinates for independent standard normal variables, gives the representation

$$rY_m \stackrel{d}{=} \frac{Z^2}{Z^2/r + W_r/r},$$

where Z is standard normal and W_r is an independent sum of r squared standard normal variables. Since $\mathbb{E}[W_r/r] = 1$ and $\text{Var}(W_r/r) = 2/r$, we obtain $rY_m \Rightarrow Z^2$. The limiting distribution is continuous and strictly increasing on $(0, \infty)$, so its upper quantile $y_\infty(v)$ satisfies $ry_m(v) \rightarrow y_\infty(v)$ for every $0 < v < 1$. Fatou's lemma therefore gives

$$\liminf_{m \rightarrow \infty} (m-1)\mathcal{J}_m \geq \int_0^1 y_\infty(v) \log(1/v) dv = \mathbb{E} \left[R^2 \log \frac{1}{Q(R)} \right], \quad (3.5)$$

where $R = |Z|$ and $Q(x) = \mathbb{P}(R > x)$. The last equality follows since $Q(R)$ is uniform on $(0,1)$. This expectation is finite: integration of the half-normal density on $[x, x+1]$ gives $Q(x) \geq \sqrt{2/\pi} e^{-(x+1)^2/2}$.

For $x > 0$, integration of the same density gives the classical Mills upper bound (see [12, Section 4, p. 113])

$$Q(x) = \sqrt{\frac{2}{\pi}} \int_x^\infty e^{-t^2/2} dt \leq \sqrt{\frac{2}{\pi}} \frac{e^{-x^2/2}}{x}.$$

Consequently, writing $A = \mathbb{E}[R^2 \log R]$, we have

$$\mathbb{E} \left[R^2 \log \frac{1}{Q(R)} \right] \geq \frac{3}{2} + A + \frac{1}{2} \log \frac{\pi}{2}. \quad (3.6)$$

Integration by parts against the half-normal density yields $\mathbb{E}[R^4 \log R] = 3A + 1$. Under the probability measure $d\nu = R^4 d\mathbb{P}/3$, Jensen's inequality for the logarithm applied to $1/R$ now gives

$$A + \frac{1}{3} = \mathbb{E}_\nu[\log R] \geq -\log \mathbb{E}_\nu[1/R] = \log \frac{3}{\mathbb{E}[R^3]} = \log \frac{3}{2\sqrt{2/\pi}}.$$

Combining this with (3.6) proves

$$\mathbb{E} \left[R^2 \log \frac{1}{Q(R)} \right] \geq \frac{7}{6} + \log \frac{3\pi}{4} > 2. \quad (3.7)$$

Here the final strict inequality can be checked without decimal approximations. Indeed, $\arctan x > x - x^3/3$ for $x > 0$ gives

$$\frac{\pi}{4} = \arctan \frac{1}{2} + \arctan \frac{1}{3} > \frac{505}{648} > \frac{7}{9},$$

so $3\pi/4 > 7/3$. Strict convexity of $t \mapsto 1/t$ gives the midpoint bounds

$$\log \frac{7}{3} = \int_1^{5/3} \frac{dt}{t} + \int_{5/3}^{7/3} \frac{dt}{t} > \frac{2/3}{4/3} + \frac{2/3}{2} = \frac{5}{6}.$$

Equations (3.5) and (3.7) imply (3.4). $\square$

Before selecting the link, fix a smooth nondecreasing cutoff χ , zero near $(-\infty, 2]$ and one near $[4, \infty)$, and put

$$I_\chi = \int_2^\infty \chi(t)(1+t)^{-5/4} dt > 0.$$

The radial construction will require only the extra smallness condition

$$\frac{-\log a_*}{I_\chi} \sup_{t>2} \chi(t)(1+t)^{-5/4} \leq \frac{1}{4}. \quad (3.8)$$

It depends on the fixed cutoff and link scale, before any finite degree range or control family has been chosen. The freedom in the next lemma allows this condition to be imposed at the outset.

Lemma 3.3 (Spectral surplus). *There is an odd $m = 2s - 1 \geq 7$ for which the following choices are possible. The parameters $a_* \in (0, 1)$ and $q_* > 1$ can be taken arbitrarily close to 1, with (2.1) holding at $a = a_*$ for every $q \in [1, q_*]$. For these parameters there are integers $k \geq 2$, $L \geq 2$, and $0 \leq M_l \leq N_l$ for $2 \leq l \leq L$ such that, on abbreviating $d_{l,i} = d_{l,i}(a_*, q_*)$,*

$$\bar{d}_l := \frac{1}{M_l} \sum_{i=1}^{M_l} d_{l,i} < k \quad \text{whenever } M_l > 0, \quad \sum_{l=2}^L M_l > \sum_{l=0}^k N_l. \quad (3.9)$$

Proof. Choosing the link. Choose an odd $m \geq 7$ satisfying Lemma 3.2, and fix

$$\frac{2}{m-1} < c < \mathcal{J}_m.$$

For a small positive parameter ε , set

$$q_* = 1 + \varepsilon, \quad \alpha_*^2 = 1 - c\varepsilon, \quad a_*^2 = q_*^{1/m} \alpha_*^2. \quad (3.10)$$

For $q \geq 1$, the right-hand side of (2.1) is

$$F(q) = q^{1/m} \left(1 - \frac{2(q-1)}{m-1} \right), \quad F'(q) = \frac{m+1}{m(m-1)} q^{1/m-1} (1-2q) < 0.$$

At $q = q_*$, the required strict inequality follows from $c > 2/(m-1)$; it consequently holds throughout $[1, q_*]$. In particular $a_*^2 < F(q_*) < 1$.

Limiting means. By (2.2), the eigenvalues at the dwell parameters are the numbers

$$\alpha_*^{-2} \left[B_l - \left(1 - \frac{1}{q_*} \right) (2b-l)^2 \right]$$

with the multiplicities in Lemma 3.1. Thus their increasing order corresponds to the decreasing order of the Hopf weights $(2b-l)^2$. If $d(d+m-1) = \lambda$ and $d \geq 0$, then $0 \leq \sqrt{\lambda} - d \leq (m-1)/2$. Since $B_l/l^2 \rightarrow 1$, Lemma 3.1 and convergence of quantiles to a continuous, strictly increasing distribution imply

$$\frac{d_{l, \lceil vN_l \rceil}}{l} \rightarrow f_\varepsilon(v) := (1 - c\varepsilon)^{-1/2} \sqrt{1 - \left(1 - \frac{1}{q_*} \right) y_m(v)} \quad (0 < v < 1). \quad (3.11)$$

These rescaled exponents are uniformly bounded, so averaging the first $\lceil uN_l \rceil$ terms gives, for every $0 < u < 1$,

$$\frac{1}{l \lceil uN_l \rceil} \sum_{i=1}^{\lceil uN_l \rceil} d_{l,i} \rightarrow g_\varepsilon(u) := \frac{1}{u} \int_0^u f_\varepsilon(v) dv. \quad (3.12)$$

One may see this directly by integrating the empirical quantile in (3.11); the possible last fractional term has size $O(N_l^{-1})$ after division by l .

Define

$$I(\varepsilon) := \int_0^1 g_\varepsilon(u)^{-m} du.$$

All functions here are bounded away from zero for sufficiently small ε . Uniformly for $0 < v < 1$,

$$f_\varepsilon(v) = 1 + \frac{\varepsilon}{2} (c - y_m(v)) + O(\varepsilon^2).$$

The same uniform remainder remains valid after averaging over $(0, u)$. Expansion of $g_\varepsilon(u)^{-m}$ and integration therefore give

$$\begin{aligned} I(\varepsilon) &= 1 + \frac{m\varepsilon}{2} \left[\int_0^1 \frac{1}{u} \int_0^u y_m(v) dv du - c \right] + O(\varepsilon^2) \\ &= 1 + \frac{m\varepsilon}{2} (\mathcal{J}_m - c) + O(\varepsilon^2) > 1 \quad \text{for all sufficiently small } \varepsilon > 0. \end{aligned} \quad (3.13)$$

The second equality uses the nonnegative integral identity $\int_v^1 du/u = \log(1/v)$. Fix such an ε , as small as desired, from now on.

Counting at integer thresholds. For each positive integer k and $l \geq 2$, let $M_l(k)$ be the largest initial group of the ordered exponents whose mean is strictly less than k , taking $M_l(k) = 0$ if no group qualifies. Only finitely many degrees contribute: the displayed eigenvalue formula bounds all $\lambda_{l,i}$ below by a positive constant times l^2 , so $\min_i d_{l,i}$ grows at least linearly in l . The initial-group means are nondecreasing, and hence

$$M_l(k) = N_l \int_0^1 \mathbf{1} \left\{ \frac{1}{\lceil u N_l \rceil} \sum_{i=1}^{\lceil u N_l \rceil} d_{l,i} < k \right\} du. \quad (3.14)$$

For fixed $u \in (0, 1)$ and $\delta > 0$, (3.12) bounds the mean in this indicator by $l(g_\varepsilon(u) + \delta/2)$ for every sufficiently large l . It is therefore strictly below k whenever $l \leq k/(g_\varepsilon(u) + \delta)$, including at the upper endpoint. Using (3.3), the sum of the corresponding N_l , divided by $\sum_{l=0}^k N_l$, has limit $(g_\varepsilon(u) + \delta)^{-m}$ as the integer $k \rightarrow \infty$. Letting $\delta \downarrow 0$ and applying Fatou's lemma to (3.14) yields

$$\liminf_{\substack{k \rightarrow \infty \\ k \in \mathbb{N}}} \frac{\sum_{l=2}^{\infty} M_l(k)}{\sum_{l=0}^k N_l} \geq \int_0^1 g_\varepsilon(u)^{-m} du = I(\varepsilon) > 1. \quad (3.15)$$

Discarding the finitely many lower degrees in this argument changes no limit. Consequently every sufficiently large integer k has a strict surplus. Taking L to be the largest degree with $M_l(k) > 0$ and setting $M_l = M_l(k)$ proves (3.9). $\square$

A selected prefix may be empty, a singleton, or all of V_l , and may cut a repeated real eigenspace. No gap at its last eigenvalue is required: choose the indicated number of real orthonormal eigenvectors. The selected span need not be invariant under any individual D_J ; the full fixed space V_l remains invariant under every angular operator.

Finally, the sum appearing in (3.9) is exactly the Euclidean comparison dimension in ambient dimension $n = m + 1$:

$$\sum_{l=0}^k N_l = \binom{m+k}{k} + \binom{m+k-1}{k-1} = h_k(\mathbb{R}^{m+1}).$$

To justify the Euclidean identification directly, let u be a Euclidean harmonic function of growth at most the integer k . A smooth radial kernel $\psi_R(x) = R^{-(m+1)}\psi(x/R)$, supported in the ball of radius R and with integral 1, reproduces u by the mean-value property. For each multi-index α of order $k+1$, moving derivatives onto the kernel gives, at any fixed x and for $R \geq 1$,

$$|\partial^\alpha u(x)| \leq \sup_{|y-x| \leq R} |u(y)| \|\partial^\alpha \psi_R\|_{L^1} \leq C_x R^k R^{-(k+1)} \rightarrow 0.$$

Thus u is a polynomial of degree at most k , and its homogeneous components are harmonic. The converse follows from homogeneity.

For the rest of the construction, the spectral data mean an odd $m = 2s - 1$ with $s \geq 4$, a fixed cutoff χ , and $0 < a_* < 1 < q_*$ satisfying (2.1) throughout $[1, q_*]$ and (3.8), together with finite $k, L \geq 2$ and $0 \leq M_l \leq N_l$ satisfying (3.9). These are precisely the numerical hypotheses used in Sections 4–7; no later step uses the asymptotic route by which the data were obtained.

The choices in Lemma 3.3 are made in the order m, c, ε, k, L . In particular any further requirement that a_* be close to 1 can be imposed when choosing ε , before the integer cutoff is fixed.

4 Control of the angular modes

The selections in Lemma 3.3 will be mixed by cycling their basis directions. We first prove that the angular operators furnish the required finite-dimensional control. Throughout this section, $m = 2s - 1$, $s \geq 4$, and $2 \leq l \leq L$. We equip each real space V_l of round spherical harmonics with its round L^2 inner product and write $N_l = \dim_{\mathbb{R}} V_l$. For an endomorphism A of V_l , put

$$A_{\text{tf}} = A - \frac{\text{tr } A}{N_l} I.$$

For every orthogonal complex structure J on $\mathbb{R}^{2s}$, the operator $D_J u(x) = du_x(Jx)$ preserves V_l . It is skew-adjoint, because its flow consists of round isometries. In particular, $(D_J^2)_{\text{tf}}$ is a real symmetric trace-free operator.

Proposition 4.1 (Simultaneous control). *Fix positive numbers κ_l , $2 \leq l \leq L$. As J ranges over all orthogonal complex structures on $\mathbb{R}^{2s}$, the tuples*

$$X_J = (\kappa_l (D_J^2|_{V_l})_{\text{tf}})_{l=2}^L \quad (4.1)$$

generate, under real linear combinations and brackets, the full Lie algebra

$$\bigoplus_{l=2}^L \mathfrak{sl}(V_l).$$

Proof. We first prove the assertion in a single degree l . The nonzero factor κ_l has no effect on this assertion. Let $\mathfrak{a} \subset \mathfrak{sl}(V_l)$ be the real Lie algebra generated by $(D_J^2)_{\text{tf}}$, and set

$$W = V_l \otimes_{\mathbb{R}} \mathbb{C}, \quad \mathfrak{a}_{\mathbb{C}} = \mathfrak{a} \otimes_{\mathbb{R}} \mathbb{C}.$$

Transpose on W will always refer to the complex bilinear extension of the real round inner product. The single-degree argument first produces a rank-one symmetric endomorphism of W . Round conjugation and brackets then produce all trace-free endomorphisms of W . Finally, we separate the different degrees to obtain simultaneous control on their direct sum.

Diagonal operators and individual weight blocks. Choose orthonormal coordinates $(x_1, y_1, \dots, x_s, y_s)$ and write $z_i = x_i + iy_i$. For $\theta \in \mathbb{R}^s$, let R_θ rotate coordinate plane i by θ_i , and let $T_\theta u(x) = u(R_\theta x)$. The *weight space* of $\nu \in \mathbb{Z}^s$ is the character eigenspace

$$W_\nu = \{u \in W : T_\theta u = e^{i \sum_i \nu_i \theta_i} u \text{ for every } \theta\}.$$

Thus z_i has weight e_i , $\bar{z}_i$ has weight $-e_i$, and D_{J_0} acts on W_ν by $i \sum_i \nu_i$. In particular, a monomial of holomorphic degree b and antiholomorphic degree $l - b$ has Hopf charge $2b - l$, consistently with Section 3. Every occurring weight satisfies

$$\sum_{i=1}^s |\nu_i| \leq l, \quad l - \sum_{i=1}^s |\nu_i| \in 2\mathbb{Z}.$$

Every weight on the extreme layer

$$\Lambda_l = \left\{ \nu \in \mathbb{Z}^s : \sum_{i=1}^s |\nu_i| = l \right\}$$

occurs and has a one-dimensional weight space: it is spanned by the monomial using $z_i^{\nu_i}$ when $\nu_i \geq 0$ and $\bar{z}_i^{-\nu_i}$ when $\nu_i < 0$. This monomial is harmonic, since in each coordinate plane it involves at most one of $z_i, \bar{z}_i$. It is the only degree- l monomial of that weight.

More generally, every weight satisfying the displayed bound and parity condition occurs. To check its multiplicity, set $a = (l - \sum_i |\nu_i|)/2$, a nonnegative integer. Every monomial of weight ν is the corresponding signed monomial of degree $\sum_i |\nu_i|$, multiplied by $\prod_i (z_i \bar{z}_i)^{a_i}$ with $a_i \geq 0$ and $\sum_i a_i = a$. The full polynomial weight space therefore has dimension $\binom{a+s-1}{s-1}$. The polynomial Laplacian preserves the weight and maps onto the corresponding weight space two degrees lower: its adjoint in the positive monomial inner product is a nonzero scalar multiple of multiplication by $\sum_i z_i \bar{z}_i$, which preserves weight and is injective. Thus the harmonic weight multiplicity is

$$\binom{a+s-1}{s-1} - \binom{a+s-2}{s-1} = \binom{a+s-2}{s-2},$$

where the second term is zero when $a = 0$. This standard weight formula also appears in Lauret–Miatello–Rossetti [10, Lemma 3.6]. In particular, interior weight spaces may have dimension greater than one. The interpolation below isolates an entire source-to-target weight block; we use a rank-one conclusion only when both endpoints are on the extreme layer.

Conjugating by a round rotation sends the family of generators to itself. Thus $\mathfrak{a}_{\mathbb{C}}$ is invariant under the round torus and contains each torus Fourier component of any of its elements. Indeed, Fourier projection is an integral of scalar multiples of torus conjugates, and a finite-dimensional vector subspace is closed.

For $\sigma = (\sigma_1, \dots, \sigma_s) \in \{\pm 1\}^s$, let J_σ rotate coordinate plane i with sign σ_i . On the ν -weight space, $D_{J_\sigma}^2$ acts by $-(\sum_i \sigma_i \nu_i)^2$. For $i \neq j$, the identity

$$2^{-s} \sum_{\sigma \in \{\pm 1\}^s} \sigma_i \sigma_j \left[- \left(\sum_a \sigma_a \nu_a \right)^2 \right] = -2\nu_i \nu_j$$

shows that $\mathfrak{a}_{\mathbb{C}}$ contains a diagonal operator A_{ij} acting on weight ν by $\nu_i \nu_j + c_{ij}$, where c_{ij} is independent of ν . The scalar term has no effect on brackets.

An endomorphism Fourier component of *weight shift* δ satisfies $T_\theta A T_\theta^{-1} = e^{i \sum_i \delta_i \theta_i} A$; equivalently, it maps each W_ν into $W_{\nu+\delta}$, with absent weight spaces understood to be zero. On its block from W_ν to $W_{\nu+\delta}$, the operator $\text{ad } A_{ij}$ acts by

$$\nu_i \delta_j + \nu_j \delta_i + \delta_i \delta_j. \tag{4.2}$$

If δ has at least three nonzero coordinates, these joint eigenvalues distinguish all source weights. In fact, equality for two sources, with difference v , implies

$$v_i \delta_j + v_j \delta_i = 0 \quad (i \neq j).$$

On the support of δ , the ratios v_i/δ_i have pairwise sums zero; three such coordinates force every ratio to be zero. Pairing any coordinate outside the support with one in the support then gives $v_i = 0$ there as well.

There are only finitely many weights. Polynomials in the commuting operators $\text{ad } A_{ij}$ therefore project onto an individual source block within a fixed shift component. More explicitly, for each competing source choose one coordinate of (4.2) that distinguishes it from the desired source, and multiply the corresponding linear interpolation factors. These operations preserve $\mathfrak{a}_{\mathbb{C}}$. When source and target belong to Λ_l , the isolated nonzero block is a rank-one directed matrix.

A rank-one symmetric operator. Regard an extreme weight as l signed tokens, with no opposite signs at the same coordinate. We claim that the algebra contains the directed matrices for every legal move

two equal-sign tokens at one coordinate $\longleftrightarrow$ one token at each of two other coordinates,

where the two latter signs are arbitrary and both endpoints must remain in Λ_l . Such a shift has three nonzero coordinates. It remains to exhibit a generator with a nonzero block for the move.

Here is an explicit simultaneous choice of the needed coefficients. For three distinct coordinate indices i, j, k , let X_a, Y_a denote their real coordinate vectors and set

$$JX_i = X_j, \quad JY_i = X_k, \quad JY_j = Y_k,$$

with the three reverse images determined by $J^2 = -I$. Use the standard complex structure on the remaining coordinate planes. This defines an orthogonal complex structure. For signed variables $u_a^\epsilon = x_a + i\epsilon y_a$, $\epsilon \in \{\pm 1\}$, its vector field satisfies

$$\begin{aligned} D_J u_i^\epsilon &= -x_j - i\epsilon x_k, \\ D_J u_j^\beta &= x_i - i\beta y_k, \\ D_J u_k^\gamma &= y_i + i\gamma y_j. \end{aligned}$$

For every choice of the three signs, the relevant replacement coefficients are

$$\begin{array}{cccc} u_i^\epsilon \longmapsto u_j^\beta & u_i^\epsilon \longmapsto u_k^\gamma & u_j^\beta \longmapsto u_i^\epsilon & u_k^\gamma \longmapsto u_i^\epsilon \\ -\frac{1}{2} & -\frac{i\epsilon}{2} & \frac{1}{2} & -\frac{i\epsilon}{2} \end{array}$$

For completeness, there is no cancellation with other terms of D_J^2 . In the independent complex linear coordinates $z_a, \bar{z}_a$, the identity $D_J^2 u = -u$ for linear functions gives, on homogeneous degree- l polynomials,

$$D_J^2 p = -lp + \sum_{u,v} (D_J u)(D_J v) \partial_u \partial_v p.$$

If

$$p = (u_i^\epsilon)^a (u_j^\beta)^b (u_k^\gamma)^c M$$

is an extreme monomial, where M uses the other coordinate planes, the coefficient of the split monomial is $i\epsilon a(a-1)/2$, and that of the merge monomial is $-i\epsilon bc/2$. They are nonzero whenever the indicated move is legal. Indeed the exponent deficits of the target force the differentiated variables to be precisely the two variables specified by the move. Since rotations commute with the Euclidean Laplacian, $D_J^2 p$ is already harmonic; no harmonic projection changes these coefficients. Trace subtraction does not affect this shift. Fourier projection and the block isolation above now give the claimed directed matrices.

We next join $-le_1$ to le_1 by a simple path of these moves. For even $l = 2h$, use

$$-2he_1 \longrightarrow -(2h-2)e_1 + e_2 + e_3 \longrightarrow \cdots \longrightarrow he_2 + he_3 \longrightarrow \cdots \longrightarrow 2he_1,$$

splitting negative pairs at coordinate 1 in the first half and merging the tokens at 2, 3 into positive pairs at 1 in the second half. This includes $l = 2$, for which the path is $-2e_1, e_2 + e_3, 2e_1$.

For odd $l = 2h + 1$, $h \geq 1$, both starting signs admit the path

$$\pm(2h+1)e_1 \longrightarrow \cdots \longrightarrow \pm e_1 + he_2 + he_3 \longrightarrow (h-1)e_2 + he_3 + 2e_4.$$

The last step merges the remaining token at 1 and one positive token at 2 into two positive tokens at 4. Reverse the positive-start path and append it to the negative-start path. The resulting path is simple: the two branches have opposite nonzero first coordinates until their common endpoint, and no branch repeats a weight. For $l = 3$, that endpoint is $e_3 + 2e_4$. All intermediate weights remain extreme.

Choose monomial bases on these one-dimensional weight spaces and normalize their directed matrices to matrix units. Nested brackets along a simple path give its endpoint unit, by $[E_{c,b}, E_{b,a}] = E_{c,a}$ for distinct vertices. Hence $\mathfrak{a}_{\mathbb{C}}$ contains the unit from weight $-le_1$ to weight le_1 . If $v = z_1^l$, torus invariance of the bilinear round pairing shows that v pairs only with weight $-le_1$, and

$$v^{\top} \bar{v} = \int_{S^{2s-1}} |z_1|^{2l} d \text{vol}_{g_0} > 0.$$

The endpoint unit is consequently a nonzero multiple of $vv^{\top}$. Notice also that $v^{\top}v = 0$.

From the rank-one operator to all trace-free matrices. All round conjugates of $vv^{\top}$ belong to the algebra. We first verify that the round orbit of v spans W . Up to a nonzero scalar, its members are the polynomials $(a \cdot x)^l$ with

$$a \in \mathbb{C}^{2s} \setminus \{0\}, \quad a \cdot a = 0.$$

To see this, write $a = b + ic$ with b, c real. The isotropy condition says that b, c are orthogonal and have equal positive length. After rescaling, this ordered pair is the image of the first coordinate pair under a real special orthogonal transformation.

On homogeneous degree- l polynomials use the Fischer bilinear form

$$(p, q)_{\text{F}} = p(\partial)q = \sum_{|\alpha|=l} \alpha! p_{\alpha} q_{\alpha}.$$

Its restriction to real harmonic polynomials is positive definite, so its complex bilinear restriction to W is nondegenerate. If $p \in W$ annihilates the orbit powers, then

$$(p, (a \cdot x)^l)_{\text{F}} = l! p(a)$$

shows that p vanishes on the isotropic quadric. Therefore p is divisible by the quadratic polynomial $Q(x) = \sum_{a=1}^s (x_a^2 + y_a^2)$. An elementary verification is to divide by this monic polynomial in the last variable y_s ; the remainder has the form $A(x')y_s + B(x')$, where x' denotes the remaining variables. Evaluating at the two distinct roots for generic x' forces both A and B to vanish identically.

Such a harmonic multiple must be zero. Indeed multiplication by Q is adjoint to the Euclidean Laplacian in the positive Hermitian Fischer form. If $p = Qq$ and $\Delta p = 0$, then

$$\|p\|_{\text{F}}^2 = \langle Qq, p \rangle_{\text{F}} = \langle q, \Delta p \rangle_{\text{F}} = 0.$$

Thus no nonzero element annihilates the orbit, proving its span.

For two orbit vectors u, w ,

$$[uu^{\top}, ww^{\top}] = (u^{\top}w)(uw^{\top} - wu^{\top}).$$

For fixed $u \neq 0$, the factor $u^{\top}w$ is not identically zero on the orbit, by its spanning property and nondegeneracy of the round pairing. It is a real-analytic function of the round rotation. The real special orthogonal group is connected, so its nonzero locus is dense. Division by this factor and

then passage to limits show that $\mathfrak{a}_{\mathbb{C}}$ contains $uw^{\top} - wu^{\top}$ for every pair of orbit vectors. Their span is the full complex skew-symmetric algebra, since the orbit spans W .

For every real skew-symmetric K , the operator $\exp(\text{ad } K)$ preserves $\mathfrak{a}_{\mathbb{C}}$; this uses only bracket closure in the complex algebra. It follows that $\mathfrak{a}_{\mathbb{C}}$ is invariant under conjugation by the full real group $SO(V_l)$. Here one may use that every special orthogonal matrix is a product of plane rotations. This group on the harmonic-function space sends v , up to a scalar, to every nonzero isotropic vector of W : their real and imaginary parts are orthogonal equal-length pairs, and $\dim V_l \geq 3$. Consequently the algebra contains $ww^{\top}$ for every isotropic $w \in W$.

These matrices span all trace-free symmetric matrices. In a real orthonormal basis, the rank-one matrices associated with $e_i + ie_j$ and $e_i - ie_j$ have sum $2(E_{ii} - E_{jj})$ and difference $2i(E_{ij} + E_{ji})$. Together with the skew-symmetric matrices already obtained, they span $\mathfrak{sl}(W)$. We have proved $\mathfrak{a}_{\mathbb{C}} = \mathfrak{sl}(W)$. Since $\mathfrak{a} \subset \mathfrak{sl}(V_l)$ is a real vector subspace and complexification preserves dimension, it follows that $\mathfrak{a} = \mathfrak{sl}(V_l)$.

Independence of the degrees. Let $\mathfrak{h}$ be the Lie algebra generated by the tuples (4.1). Every one-factor projection is full by what we have just proved. Moreover

$$N_l = \binom{m+l}{m} - \binom{m+l-2}{m} = \binom{m+l-1}{m-1} + \binom{m+l-2}{m-1}$$

is strictly increasing in l . The real Lie algebras $\mathfrak{sl}(V_l)$ are therefore simple and pairwise nonisomorphic. Simplicity can be seen directly: in any nonzero ideal, a bracket with a matrix unit first produces an element with an off-diagonal entry if necessary; polynomials in the adjoint action of a generic trace-free diagonal matrix isolate an off-diagonal unit. Brackets with the other matrix units then produce all off-diagonal units and all diagonal differences.

We give the resulting Lie-algebra Goursat argument explicitly; for the two-factor formulation, see [7, Section 3]. Inductively, suppose the projection onto a preceding product $\mathfrak{b}$ is full, and let $\mathfrak{c}$ be the next simple factor. The projected algebra $\mathfrak{h}' \subset \mathfrak{b} \oplus \mathfrak{c}$ is surjective onto both sides. The kernel of its projection onto $\mathfrak{b}$ is an ideal of $\mathfrak{c}$, and hence is either zero or $\mathfrak{c}$. In the latter case the whole product is present. In the former case $\mathfrak{h}'$ is the graph of a surjective homomorphism $\mathfrak{b} \rightarrow \mathfrak{c}$. The image of each simple summand of $\mathfrak{b}$ is an ideal in $\mathfrak{c}$; a nonzero image would make that summand isomorphic to $\mathfrak{c}$. This contradicts their distinct dimensions. The induction proves the claimed full direct sum. $\square$

Remark 4.2. The quantifier over all orthogonal complex structures in Proposition 4.1 includes both orientation classes. This is used in the averaging over all sign vectors. In particular, when $s = 4$, restricting to one class would identify some complementary sign characters and would not justify that averaging step. The explicit complex structure used for the token moves can itself be completed in either class by reversing it on an unused coordinate plane.

For general background on control systems on Lie groups, see Jurdjevic–Sussmann [9]. We give the finite-word argument with its nonsingular differential directly.

We now pass from the Lie algebra to exact finite products. For the integers M_l in (3.9), choose the ordered real orthonormal dwell eigenbasis $e_{l,1}, \dots, e_{l,N_l}$. If $M_l \geq 1$, define a signed cycle by

$$\Pi_l e_{l,i} = \begin{cases} e_{l,i+1}, & 1 \leq i < M_l, \\ (-1)^{M_l-1} e_{l,1}, & i = M_l, \\ e_{l,i}, & i > M_l, \end{cases}$$

and set $\Pi_l = I$ when $M_l = 0$. The sign of the M_l -cycle and its extra column sign cancel, so $\det \Pi_l = 1$. Thus

$$\Pi = (\Pi_l)_{l=2}^L \in \mathcal{G} := \prod_{l=2}^L SL(V_l).$$

Corollary 4.3 (A finite word with invertible differential). *Let $d = \sum_{l=2}^L (N_l^2 - 1)$. There exist finitely many orthogonal complex structures $J_1, \dots, J_R$, an open neighborhood U of zero in $\mathbb{R}^d$, and smooth real functions $c_1, \dots, c_R$ on U such that*

$$\mathcal{W}(h) = \exp(c_R(h)X_{J_R}) \cdots \exp(c_1(h)X_{J_1})$$

satisfies $\mathcal{W}(0) = \Pi$, and

$$D\mathcal{W}(0) : \mathbb{R}^d \longrightarrow T_\Pi \mathcal{G}$$

is an isomorphism. The same assertion holds for any prescribed target in $\mathcal{G}$.

Proof. Let $\mathcal{H}$ be the subgroup of $\mathcal{G}$ consisting of finite products of $\exp(tX_J)$ with arbitrary real t . Define

$$E = \text{span}_{\mathbb{R}} \{\text{Ad}_h X_J : h \in \mathcal{H}\}.$$

Conjugation by every generator exponential preserves E . Differentiating shows that E is stable under bracketing with each X_J . Since E contains the generators, it contains their generated Lie algebra, which is the whole Lie algebra of $\mathcal{G}$ by Proposition 4.1. Choose a basis $\text{Ad}_{h_a} X_{K_a}$, $1 \leq a \leq d$, from this spanning set. The map

$$F(t_1, \dots, t_d) = (h_d \exp(t_d X_{K_d}) h_d^{-1}) \cdots (h_1 \exp(t_1 X_{K_1}) h_1^{-1})$$

has value I at zero and differential

$$DF(0)(t) = \sum_{a=1}^d t_a \text{Ad}_{h_a} X_{K_a},$$

which is an isomorphism. Every factor h_a , and its inverse, is a fixed finite word of generator exponentials. Thus every value of F is in $\mathcal{H}$.

The inverse function theorem gives an identity neighborhood contained in $\mathcal{H}$, making $\mathcal{H}$ an open subgroup. The group $SL_N(\mathbb{R})$ is connected: polar decomposition reduces this to connectedness of $SO(N)$ and of the positive definite determinant-one matrices, the latter being joined to the identity by S^t , $0 \leq t \leq 1$. Hence $\mathcal{G}$ is connected, and its open subgroup $\mathcal{H}$ is all of $\mathcal{G}$.

Choose a fixed finite word representing the desired target and multiply it by F . The resulting word has that target as its value at zero and still has invertible differential. Expanding the fixed conjugating words gives the asserted form, with fixed complex structures and smooth real time parameters. $\square$

Remark 4.4. The real times in Corollary 4.3 may have either sign. They will be realized as integrals of bounded smooth bumps z in weak metric pulses of the form $q = 1 + z/T$. Both signs are admissible once T is sufficiently large, because $q > 0$ and converges uniformly to 1. Likewise, the class of J may be changed while $q = 1$, when the angular metric is independent of J . Thus neither the signed times nor the two orientation classes impose a restriction on the later construction.

5 A complete metric with nonnegative Ricci curvature

Fix spectral data in the sense specified at the end of Section 3. The geometric properties we need are $0 < a_* < 1 < q_*$, the strict inequality (2.1) at $a = a_*$ for every $q \in [1, q_*]$, and the fixed cutoff condition (3.8). Fix also the reference complex structure J_0 used for the dwell metric. We describe a metric construction that works for any prescribed finite family of pulse controls. Its curvature estimates will be uniform over all subsequent choices of those controls. Slowly varying links with fixed volume form also appear in Colding–Naber [5, Lemma 2.1]. We give the curvature calculation for the present family in full.

The radial cutoff and link parameters were fixed before k, L and before any pulse family. Condition (3.8) will give $b = a'/a \geq -1/4$ below. Even very large bounds for a fixed finite pulse family are handled solely by increasing the starting index j_0 . No spectral parameter is changed in response to the controls.

Pulse data and the period schedule. Let K be a compact ball in a finite-dimensional real parameter space. Choose finitely many disjoint open subintervals $I_1, \dots, I_R$ of $(0, 1)$, with disjoint closures, smooth functions β_ν compactly supported in I_ν , and orthogonal complex structures J_ν . For each ν , let $c_\nu(h)$ be smooth on a neighborhood of K . Set

$$z(\tau, h) = \sum_{\nu=1}^R c_\nu(h) \beta_\nu(\tau).$$

The amplitudes c_ν may have either sign. The functions, complex structures, and compact set K are fixed once and for all. In applications the bumps can be normalized to have integral 1, so their amplitudes specify the flow parameters in Corollary 4.3.

For $j \geq 1$, define

$$T_j = j^6, \quad L_j = j^7, \quad t_1 = 10, \quad t_{j+1} = t_j + L_j + 4T_j. \quad (5.1)$$

Choose a starting index j_0 , to be made large below. Put $q(t) = 1$ for $t \leq t_{j_0}$. For each $j \geq j_0$, choose a parameter $h_j \in K$ and divide $[t_j, t_{j+1}]$ into the following five successive intervals:

1. a pulse of duration T_j , on which

$$q(t) = 1 + T_j^{-1} z((t - t_j)/T_j, h_j);$$

on the support of the ν -th bump the complex structure is J_ν ;

2. an upward ramp of duration T_j , taking q from 1 to q_* ;
3. a dwell interval of duration L_j , on which $q = q_*$;
4. a downward ramp of duration T_j , taking q from q_* to 1;
5. a round rest of duration T_j , on which $q = 1$.

Use J_0 throughout the last four intervals. Both ramps use a fixed smooth nondecreasing profile $\rho : [0, 1] \rightarrow [0, 1]$, equal to 0 near 0 and 1 near 1: their respective formulas in their own rescaled time are $1 + (q_* - 1)\rho$ and $q_* - (q_* - 1)\rho$. The last four intervals form the diagonal leg of a period, as illustrated in Figure 1.

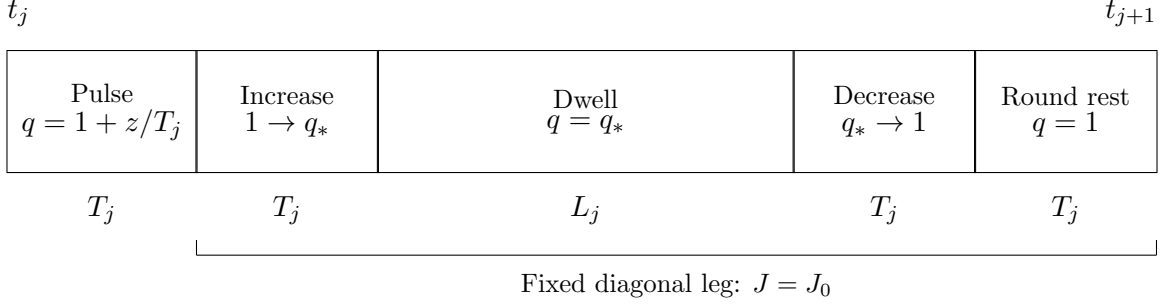


Figure 1: A period in logarithmic radius, with $T_j = j^6$ and $L_j = j^7$. The pulse contains finitely many weak bumps with round gaps between them. The four subsequent pieces form the fixed diagonal leg. Horizontal lengths are schematic.

The notation $J(t)$ is only a label for the complex structure used in the metric at time t . It need not be continuous. Labels are changed only in open intervals where $q = 1$, and $G(a, 1, J) = a^2 g_0$ is independent of J there. The supported bumps provide such round gaps before, between, and after their supports.

Proposition 5.1. *For the fixed spectral data above, there are a smooth function $a : \mathbb{R} \rightarrow (0, 1]$ and an index j_{geom} such that, for every $j_0 \geq j_{\text{geom}}$, every sequence $(h_j)_{j \geq j_0}$ in K , used in the preceding scheme, defines a smooth complete metric on $\mathbb{R}^{m+1}$ with nonnegative Ricci curvature:*

$$g = dr^2 + r^2 \gamma(\log r), \quad \gamma(t) = G(a(t), q(t), J(t)).$$

The function a is identically 1 for $t \leq 2$, decreases to a_* , and, for some $\mu > 0$, satisfies

$$b(t) := \frac{a'(t)}{a(t)} = -\mu(1+t)^{-5/4} \quad (t \geq 4), \quad \log \frac{a(t)}{a_*} = 4\mu(1+t)^{-1/4} \quad (t \geq 4). \quad (5.2)$$

The metric is Euclidean near the origin, and its distance from the origin is exactly r . Its volume form is

$$d \text{vol}_g = r^m a(\log r)^m dr d \text{vol}_{g_0}.$$

The function a and the starting index j_0 can be chosen independently of the sequence of control parameters.

Proof. The radial scale. Use the fixed smooth nondecreasing cutoff χ from Section 3. For $t > 2$, put

$$b(t) = -\mu \chi(t)(1+t)^{-5/4}, \quad a(t) = \exp \left(\int_2^t b(u) du \right),$$

and set $b = 0$, $a = 1$ for $t \leq 2$. The integral of $\chi(t)(1+t)^{-5/4}$ over $(2, \infty)$ is positive and finite. Thus there is a unique $\mu > 0$ for which $a(t) \rightarrow a_*$; moreover $\mu \rightarrow 0$ as $a_* \rightarrow 1$. Condition (3.8) gives $b \geq -1/4$. Since $\chi' \geq 0$, for $t > 2$ we have

$$b'(t) \leq -\frac{5b(t)}{4(1+t)}, \quad b + b' + b^2 \leq b \left(1 - \frac{5}{4(1+t)} + b \right) \leq 0.$$

The last factor is positive for $t > 2$ and $b \geq -1/4$. The tail identities (5.2) follow by integration.

The slice shape and the mixed Ricci tensor. Write $g_r = r^2\gamma(\log r)$ for the metric on a slice. A dot will denote differentiation in $t = \log r$. Identify the tangent bundles of the slices using the product coordinates. Their shape endomorphism is

$$S = \frac{1}{2}g_r^{-1}\partial_r g_r = r^{-1}(I + B), \quad B = \frac{1}{2}\gamma^{-1}\dot{\gamma} = bI + H.$$

Where a nonround part of a bump or a ramp occurs, J is fixed. Direct differentiation of G gives

$$H = \frac{\dot{q}}{2q} \left(P_J - \frac{1}{m}I \right), \quad \text{tr } H = 0, \quad \text{tr}(H^2) = \frac{m-1}{4m} \left(\frac{\dot{q}}{q} \right)^2. \quad (5.3)$$

On round gaps $H = 0$. Thus the same identities apply everywhere, without requiring derivatives of the labels $J(t)$.

The projection P_J is also the orthogonal projection for γ onto the Hopf direction. Indeed, ξ_J has constant squared length $a^2 q^{1-1/m}$ on a fixed slice, and its metric dual is the same constant times η_J . The flow of ξ_J preserves both g_0 and η_J , so ξ_J is a Killing field for γ . A Killing field of constant length has zero divergence and $\nabla_{\xi_J}^\gamma \xi_J = 0$: the latter identity follows by pairing the Killing equation with ξ_J . Equivalently, its unit normalization E satisfies

$$\text{div}_\gamma(E \otimes E^\flat) = (\text{div}_\gamma E)E^\flat + (\nabla_E^\gamma E)^\flat = 0.$$

Therefore $\text{div}_\gamma P_J = 0$, and $\text{div}_\gamma H = 0$, because the coefficients in (5.3) are constant on a slice. The trace $\text{tr } S = r^{-1}m(1+b)$ is also spatially constant. The Codazzi formula consequently makes the mixed Ricci tensor identically zero.

We record the remaining curvature identities explicitly. For a metric $dr^2 + g_r$, the connection components involving the radial variable are $\Gamma_{ij}^r = -(g_r)_{ik}S^k_j$ and $\Gamma_{rj}^i = S^i_j$. They give the normal curvature endomorphism $-\partial_r S - S^2$. Its trace is $\text{Ric}_g(\partial_r, \partial_r)$. Tracing the tangential Gauss equation contributes $S^2 - (\text{tr } S)S$ to the intrinsic slice Ricci endomorphism. Adding the normal contribution gives

$$\text{Ric}_g^\#|_{TS^m} = \text{Ric}_{g_r}^\# - \partial_r S - (\text{tr } S)S.$$

Here an index of the tangential tensor is raised using g_r . The mixed formula is $\text{div}_{g_r} S - d(\text{tr } S)$, already shown to vanish. Substitution of $S = r^{-1}(I + B)$, and $\text{tr } B = mb$, yields

$$\begin{aligned} r^2 \text{Ric}_g(\partial_r, \partial_r) &= -m(b + \dot{b} + b^2) - \text{tr}(H^2), \\ r^2 \text{Ric}_g^\#|_{TS^m} &= \text{Ric}_\gamma^\# - \dot{B} - (m(1+b) - 1)(I + B). \end{aligned} \quad (5.4)$$

In particular, the H^2 terms cancel from the tangential identity.

Uniform positivity on the periods. All constants in the estimates that follow may depend on the fixed pulse data and K , but not on j or on any choices of h_j . On a pulse,

$$|q - 1| = O(T_j^{-1}), \quad |\dot{q}| = O(T_j^{-2}), \quad |\ddot{q}| = O(T_j^{-3}).$$

On either ramp, $|\dot{q}| = O(T_j^{-1})$ and $|\ddot{q}| = O(T_j^{-2})$. There is no variation of q during the dwell or rest. Taking j_0 large makes $q > 0$ throughout all pulses, including those with negative amplitudes. The metrics on all slices then lie in a fixed uniformly equivalent family. Formula (5.3) gives, throughout period j ,

$$\|H\| = O(T_j^{-1}), \quad \|\dot{H}\| = O(T_j^{-2}), \quad \text{tr}(H^2) = O(T_j^{-2}) = O(j^{-12}).$$

The stronger pulse bound is $\text{tr}(H^2) = O(T_j^{-4})$. To obtain the derivative estimate, note that P_J is independent of t where q varies; on every label-changing gap H vanishes identically.

From (5.1),

$$t_j \sim \frac{j^8}{8}, \quad \sup_{t \in [t_j, t_{j+1}]} \left| \frac{t}{t_j} - 1 \right| \rightarrow 0.$$

On the scale tail, an exact calculation gives

$$-m(b + \dot{b} + b^2) = m\mu(1+t)^{-5/4} \left(1 - \frac{5}{4(1+t)} - \mu(1+t)^{-5/4} \right).$$

It has a positive lower bound of order j^{-10} , uniformly on period j , for all sufficiently large j . This dominates the $O(j^{-12})$ term $\text{tr}(H^2)$. The radial component in (5.4) is therefore positive once j_0 is large enough.

For the tangential component, the strict inequality (2.1) on the compact interval $[1, q_*]$ implies that, for some fixed $\delta > 0$,

$$\text{Ric}_{G(a_*, q, J_0)}^\# \geq (m - 1 + 2\delta)I \quad (1 \leq q \leq q_*).$$

All J give the same intrinsic Ricci eigenvalues. Since $a(t) \rightarrow a_*$, this estimate persists with 2δ replaced by δ along sufficiently late ramps, dwells, and round rests. The pulse values satisfy $q \rightarrow 1$ uniformly; the strict inequality at $(a_*, 1)$ likewise supplies the same conclusion on late pulses, after decreasing δ if necessary. This argument allows $q < 1$ on a pulse. Thus all sufficiently late slices satisfy

$$\text{Ric}_\gamma^\# \geq (m - 1 + \delta)I.$$

On the other hand, $B = bI + H$, $b, \dot{b} \rightarrow 0$, and the displayed bounds for $H, \dot{H}$ show uniformly that

$$\dot{B} + (m(1+b) - 1)(I + B) = (m - 1)I + o(1).$$

The tangential component of (5.4) is consequently positive as well. All the thresholds used here depend only on the fixed data and K . Since the mixed component is zero, these estimates prove $\text{Ric}_g \geq 0$ on every period for every admitted parameter sequence.

The initial region and the global manifold. Before t_{j_0} the metric is round in the angular variable:

$$g = dr^2 + f(r)^2 g_0, \quad f(r) = r a(\log r).$$

The scale construction gives

$$f'(r) = a(1+b) \in (0, 1], \quad f''(r) = \frac{a}{r}(b + \dot{b} + b^2) \leq 0.$$

Its radial and tangential sectional curvatures are respectively $-f''/f$ and $(1 - (f')^2)/f^2$, as follows from the same normal curvature and Gauss formulas. They are nonnegative, proving the required Ricci condition in this region.

The pulse bumps and ramp profiles are constant or flat at all junctions, and label changes occur only in round open intervals. Thus γ , and hence g , is smooth for $r > 0$. Only finitely many periods meet any bounded interval of t , so the infinite sequence of periods introduces no finite accumulation of junctions. For $t \leq 2$, equivalently $r \leq e^2$, both a and q equal 1. The metric is exactly Euclidean there and extends smoothly across the origin. The volume formula follows from Lemma 2.1.

A radial path from the origin to a point of radius r has length r . Conversely, the length of any such path is at least the total variation of its radial coordinate, and hence at least r . Thus $d_g(0, x) = r(x)$. The closed centered balls are the compact sets $\{r \leq R\}$ in the underlying $\mathbb{R}^{m+1}$, so the metric is complete. The underlying manifold is connected, smooth, and without boundary. $\square$

6 Exact transmission on the regular harmonic subspaces

The link, selection, and control results are now proved. Fix finite spectral data as specified at the end of Section 3; Lemma 3.3 supplies such data. Thus the real spaces V_l , the finite range $2 \leq l \leq L$, and the selected multiplicities M_l are fixed. Keep the ordered dwell eigenbases and signed cycles Π_l chosen in Section 4. Every matrix in this section acts in those fixed round orthonormal coordinates.

Our task is to make the harmonic equation realize these cycles after an entire period, including its long nonround dwell. First we choose the pulse family using Corollary 4.3. Proposition 5.1 then supplies the metric for every admitted sequence once the start is sufficiently late. The remainder of this section proves uniform estimates for those provisional metrics before choosing the actual sequence.

6.1 The ideal pulse and its geometric realization

We now choose the constants in the control proposition to agree with the response of the radial equation. Let $p_\infty = m - 1$, and for $2 \leq l \leq L$ define

$$\theta_l > 0, \quad \theta_l^2 + p_\infty \theta_l = a_*^{-2} B_l, \quad \kappa_l = \frac{a_*^{-2}}{p_\infty + 2\theta_l} > 0. \quad (6.1)$$

Apply Corollary 4.3 to the target Π with these κ_l . Restrict its parameter domain to a closed ball K centered at zero and contained in U . Choose open intervals $I_1, \dots, I_R \subset (0, 1)$, in chronological order and with disjoint closures, and smooth functions β_ν compactly supported in I_ν with $\int_0^1 \beta_\nu(\tau) d\tau = 1$. Put

$$\begin{aligned} z(\tau, h) &= \sum_{\nu=1}^R c_\nu(h) \beta_\nu(\tau), \\ R_l(\tau, h) &= \kappa_l \sum_{\nu=1}^R c_\nu(h) \beta_\nu(\tau) \left(\frac{B_l}{m} I + D_{J_\nu}^2 |_{V_l} \right). \end{aligned} \quad (6.2)$$

These functions, with all derivatives that occur below, are bounded on the fixed compact parameter set. Let H_l solve

$$\partial_\tau H_l = R_l(\tau, h) H_l, \quad H_l(0, h) = I. \quad (6.3)$$

For a matrix C on V_l with positive determinant, define

$$\mathcal{N}_l(C) = (\det C)^{-1/N_l} C.$$

Each bump in (6.2) is a scalar function of time times one constant matrix. It therefore contributes $\exp(c_\nu(h) \kappa_l (B_l I / m + D_{J_\nu}^2))$ to $H_l(1, h)$, with later bumps multiplying on the left. For positive-determinant matrices, $\mathcal{N}_l(C_2 C_1) = \mathcal{N}_l(C_2) \mathcal{N}_l(C_1)$; also $\mathcal{N}_l(e^C) = e^{C_{\text{tr}}}$ for every real matrix C . Thus removing the determinant removes the scalar part of each exponent. Consequently

$$\begin{aligned} \Phi(h) &:= (\mathcal{N}_l(H_l(1, h)))_{l=2}^L = \mathcal{W}(h), \\ \Phi(0) &= \Pi, \quad D\Phi(0) \text{ is an isomorphism.} \end{aligned} \quad (6.4)$$

The determinants of the H_l are positive, since they are exponentials of integrals of real traces.

Apply Proposition 5.1 to this fixed pulse family. Its proof supplies a threshold j_{geom} such that every $j_0 \geq j_{\text{geom}}$ and every parameter sequence in K give a smooth complete metric with nonnegative Ricci curvature. Use exactly the radial scale a , cutoff, and five-part schedule (5.1) fixed there. In particular, $d_g(0, x) = r(x)$, $b = a'/a \geq -1/4$, and the angular volume density is $a(t)^m$ times round volume. The four pieces after the pulse form the whole diagonal leg shown in Figure 1.

6.2 The center-regular value equation

Before analyzing the harmonic equation, we record one estimate for matrix differences. It does not require the matrices in a product to commute. All matrix norms in its applications are Frobenius norms from the fixed round inner products; in fixed finite dimensions they also control operator norms.

Lemma 6.1 (A damped matrix difference). *Let $E : [u, v] \rightarrow \mathbb{R}^{N \times N}$ be continuously differentiable, and let F, L, R be continuous matrix-valued functions on the same interval. Suppose*

$$E' = F - LE - ER$$

there, where L, R are symmetric and $\lambda_{\min}(L) + \lambda_{\min}(R) \geq \gamma > 0$. Then, for $u \leq t \leq v$,

$$\|E(t)\| \leq e^{-\gamma(t-u)} \|E(u)\| + \int_u^t e^{-\gamma(t-w)} \|F(w)\| dw, \quad (6.5)$$

$$\int_u^v \|E(t)\| dt \leq \gamma^{-1} \left(\|E(u)\| + \int_u^v \|F(t)\| dt \right). \quad (6.6)$$

Proof. The Frobenius inner product gives

$$\frac{1}{2} \frac{d}{dt} \|E\|^2 = \langle E, F \rangle - \langle E, LE \rangle - \langle E, ER \rangle \leq \|E\| \|F\| - \gamma \|E\|^2.$$

The corresponding upper differential inequality for $\|E\|$ holds also at its zeros, by continuity (or by first replacing the norm by $(\|E\|^2 + \varepsilon^2)^{1/2}$ and letting $\varepsilon \downarrow 0$). Multiplication by $e^{\gamma t}$ and integration gives (6.5). Integrating that estimate in t and interchanging the nonnegative integrals gives (6.6). $\square$

For a harmonic function whose angular part belongs to V_l , write

$$u(r, \omega) = \sum_{i=1}^{N_l} Y_i(\log r) e_{l,i}(\omega).$$

We identify its coefficient vector $Y(t)$ with $\sum_i Y_i(t) e_{l,i} \in V_l$. The function u is called *center-regular* if it extends smoothly across the origin. The volume formula in Proposition 5.1 is the reason that a closed equation holds on this fixed coefficient space. With $t = \log r$, it is

$$Y'' + p(t)Y' - A_l(t)Y = 0, \quad p(t) = m - 1 + mb(t), \quad A_l(t) = A_l(a(t), q(t), J(t)). \quad (6.7)$$

Indeed, the radial density is $r^m a(\log r)^m$. Its logarithmic derivative contributes $m + mb(t)$ in the radial equation, and the change $t = \log r$ subtracts 1, giving $p(t)$. The density is constant in the angular variable, and (2.2) preserves V_l , so no additional angular terms or other harmonic degrees occur.

Lemma 6.2 (The center-regular value equation). *There are a threshold $j_{\text{reg}} \geq j_{\text{geom}}$ and constants $0 < c_P < C_P$ such that the following holds for every $j_0 \geq j_{\text{reg}}$, every sequence of parameters in K , and every $2 \leq l \leq L$. There is a unique smooth symmetric matrix $P_l(t)$ satisfying*

$$P'_l = A_l - pP_l - P_l^2, \quad P_l = lI \quad (t \leq 2). \quad (6.8)$$

It satisfies

$$c_P I \leq P_l(t) \leq C_P I \quad (t \in \mathbb{R}, 2 \leq l \leq L). \quad (6.9)$$

The solutions of $Y' = P_l Y$ are exactly the center-regular solutions of (6.7). Their value vector at any one time can be prescribed arbitrarily and determines the solution uniquely. At each pulse start one also has

$$\|P_l(t_j) - \theta_l I\| \leq Cj^{-2} \quad (j \geq j_0), \quad (6.10)$$

including the first pulse. The constants and the threshold are uniform over all preceding choices of the parameters in K .

Proof. The matrices $A_l(t)$ are symmetric. For sufficiently large allowed j_0 , the values of a and q lie in fixed compact positive intervals, and only finitely many V_l are involved. Positivity of (2.2) therefore gives constants

$$0 < \lambda_- I \leq A_l(t) \leq \lambda_+ I.$$

The radial condition $b \geq -1/4$, together with $b \leq 0$, gives

$$0 < p_- := 3m/4 - 1 \leq p(t) \leq p_+ := m - 1.$$

Choose $c_P > 0$ below all the initial values l so that $\lambda_- - p_+ c_P - c_P^2 > 0$, and choose C_P above all of them so that $\lambda_+ - p_- C_P - C_P^2 < 0$. We may decrease c_P and increase C_P so that every θ_l also lies strictly between them.

The Riccati equation preserves symmetry. At a first contact with the lower barrier, a unit vector v with $P_l v = c_P v$ satisfies

$$v^\top P'_l v \geq \lambda_- - p_+ c_P - c_P^2 > 0.$$

At a contact with the upper barrier the corresponding derivative is at most $\lambda_+ - p_- C_P - C_P^2 < 0$. These strict inward derivatives prevent an eigenvalue from crossing either barrier. The bounds also keep P_l in a compact set of matrices on every finite time interval, so local existence for the ordinary differential equation continues for all later times. Uniqueness follows from the same local ordinary differential equation theorem. In the Euclidean region, $A_l = B_l I$ and $p = m - 1$, so $P_l = lI$ indeed satisfies the equation there.

Let $\mathcal{F}_l(t)$ be the fundamental matrix of $Y' = P_l Y$ with $\mathcal{F}_l(2) = I$. Its determinant is

$$\det \mathcal{F}_l(t) = \exp \left(\int_2^t \text{tr } P_l(w) dw \right) > 0.$$

Moreover,

$$Y'' = (P'_l + P_l^2)Y = A_l Y - pY',$$

so its columns solve (6.7). On $t \leq 2$ they are $e^{l(t-2)}$ times constant vectors. Conversely, the two scalar radial factors for (6.7) in this region are e^{lt} and $e^{-(l+m-1)t}$. Smoothness at the origin eliminates the second factor. Thus every center-regular solution is one of the solutions $Y' = P_l Y$. Invertibility of $\mathcal{F}_l(t)$ gives the assertion about arbitrary value data.

It remains to prove the reset estimate. The schedule gives

$$t_j = \frac{j^8}{8} + O(j^7), \quad \sup_{t \in [t_j, t_{j+1}]} \left| \frac{t}{t_j} - 1 \right| \rightarrow 0. \quad (6.11)$$

Times on period j and on the final rest immediately preceding it are therefore comparable to j^8 . From (5.2),

$$a(t) - a_* = O(j^{-2}), \quad p(t) - p_\infty = O(j^{-10}) \quad (6.12)$$

uniformly on these intervals. On a round interval set $E_0 = P_l - \theta_l I$. Subtracting (6.1) from (6.8) gives the exact difference equation

$$E'_0 = f_l^{\text{rd}}(t)I - (pI + P_l)E_0 - E_0\theta_l I, \quad f_l^{\text{rd}}(t) = a(t)^{-2}B_l - p(t)\theta_l - \theta_l^2.$$

Here $f_l^{\text{rd}} = O(j^{-2})$ by (6.12). The symmetric coefficients $pI + P_l$ and $\theta_l I$ have a fixed positive sum of lower eigenvalue bounds. Lemma 6.1, on a final rest of length T_{j-1} , gives

$$\|P_l(t_j) - \theta_l I\| \leq Ce^{-\gamma T_{j-1}} + Cj^{-2} = O(j^{-2}).$$

The discrepancy at the start of the rest is bounded by (6.9), independently of the history. For the first pulse, the interval up to t_{j_0} is entirely round. Apply the same estimate on $[t_{j_0}/2, t_{j_0}]$ once j_0 is large. Its length tends to infinity and all its times are comparable to j_0^8 , so the same bound follows. This proves (6.10) with the claimed uniformity. $\square$

6.3 The response of one pulse

The reset brings the regular radial derivative close to the round value θ_l , but its $O(j^{-2})$ error cannot simply be multiplied by the pulse duration $T_j = j^6$. The next proof uses an exact scalar radial solution for the long common growth and integrates the remaining, damped matrix error. This is the point at which the denominator $p_\infty + 2\theta_l$ in (6.1) enters. Contraction properties of positive matrix Riccati flows are classical; see Lawson–Lim [11, Theorem 8.5]. We prove the forced estimates, including their parameter derivatives, in the form used here.

Fix a period j , hold all parameters in earlier periods fixed, and let the current parameter h vary in K . Write

$$T = T_j, \quad \tau = (t - t_j)/T, \quad \eta_j = j^{-2} + T^{-1}.$$

Let $\mathcal{B}_{l,j}(h)$ be the value transmission for $Y' = P_l Y$ from the start to the end of the pulse, and write $P_{l,e}(h) = P_l(t_j + T, h)$. Define the scalar solution

$$v'_l = a(t)^{-2}B_l - p(t)v_l - v_l^2, \quad v_l(t_j) = \theta_l, \quad \varphi_{l,j} = \int_{t_j}^{t_j+T} v_l(t) dt. \quad (6.13)$$

This solution is independent of h and of the earlier parameters.

Lemma 6.3 (Pulse limit with one parameter derivative). *For the fixed compact ball and pulse family,*

$$\begin{aligned} e^{-\varphi_{l,j}} \mathcal{B}_{l,j}(h) &= H_l(1, h) + O_{C^1}(\eta_j), \\ P_{l,e}(h) &= \theta_l I + O_{C^1}(\eta_j). \end{aligned} \quad (6.14)$$

Here $O_{C^1}(\eta_j)$ means that the norm and the norm of its first derivative in the current parameter h , uniformly on K , are at most $C\eta_j$. The constants and the large- j threshold are independent of every admitted preceding history.

Proof. The scalar barriers used in Lemma 6.2 also keep v_l between fixed positive constants. Subtracting (6.1) from (6.13) gives

$$(v_l - \theta_l)' = (a(t)^{-2}B_l - p(t)\theta_l - \theta_l^2) - (p + v_l + \theta_l)(v_l - \theta_l).$$

Its initial value is zero. The forcing is $O(j^{-2})$ by (6.12), and the damping coefficient is uniformly positive. Thus

$$v_l - \theta_l = O(j^{-2}) \quad \text{throughout the pulse.} \quad (6.15)$$

We retain v_l exactly in the value equation: even the small pointwise difference in (6.15) can have a large integral over T_j .

For the angular operator, expand (2.2) at $q = 1$. Its first derivative in q is

$$a^{-2} \left(\frac{B_l}{m} I + D_J^2|_{V_l} \right).$$

In particular the trace-free part has the positive coefficient $a^{-2}(D_J^2)_{\text{tf}}$ that appears in (4.1). Since $q = 1 + z/T$ on the pulse and $a(t) - a_* = O(j^{-2})$, Taylor's formula gives

$$\begin{aligned} A_l(t, h) &= a(t)^{-2}B_l I + \frac{p_\infty + 2\theta_l}{T} R_l(\tau, h) + \mathcal{E}_{A,l}(t, h), \\ \|\mathcal{E}_{A,l}\|_{C_h^1} &\leq C \left(\frac{j^{-2}}{T} + \frac{1}{T^2} \right). \end{aligned} \quad (6.16)$$

At most one bump is nonzero at any time. In the gaps the angular operator is exactly scalar. Smooth boundedness of the profiles and their first h -derivatives proves the uniform remainder bound, also across those gaps.

Set

$$Q_l(t, h) = v_l(t)I + \frac{1}{T}R_l(\tau, h).$$

It is symmetric, and it has a fixed positive lower eigenvalue bound when j is large. Its Riccati residual is

$$\rho_l = A_l - pQ_l - Q_l^2 - Q_l'.$$

The scalar terms in this expression cancel by (6.13). Using $\partial_t(R_l/T) = \partial_\tau R_l/T^2$, the remaining identity is

$$\rho_l = -\frac{p - p_\infty + 2(v_l - \theta_l)}{T} R_l - \frac{R_l^2 + \partial_\tau R_l}{T^2} + \mathcal{E}_{A,l}.$$

Equations (6.15) and (6.16) therefore imply

$$\sup_{t,h} (\|\rho_l\| + \|D_h \rho_l\|) \leq \frac{C\eta_j}{T}. \quad (6.17)$$

Let $E_l = P_l - Q_l$. No matrices are commuted in the exact difference equation

$$E_l' = \rho_l - (pI + P_l)E_l - E_l Q_l. \quad (6.18)$$

The lower eigenvalue bounds for P_l, Q_l , and p allow Lemma 6.1 with a fixed damping rate. The profiles vanish near $\tau = 0$, so

$$E_l(t_j) = P_l(t_j) - \theta_l I = O(j^{-2})$$

by (6.10). From (6.5), (6.6), and (6.17), we obtain

$$\sup_{t,h} \|E_l(t,h)\| \leq C\eta_j, \quad \sup_h \int_{t_j}^{t_j+T} \|E_l(t,h)\| dt \leq C\eta_j. \quad (6.19)$$

The initial discrepancy contributes its size to this integral, because it is damped; it does not contribute its size times T .

We next control the derivative in a current parameter. Derivatives are taken with the earlier history fixed. Write $E_{l,h} = D_h E_l$ and $Q_{l,h} = D_h Q_l$; the calculation applies to every unit direction in the finite parameter space. The coefficients of the Riccati equation depend smoothly on h on the pulse, so the ordinary differential equation has smooth parameter dependence there; the uniform barriers keep its solutions inside one fixed bounded set. Differentiating (6.18) and using $D_h P_l = E_{l,h} + Q_{l,h}$ gives the exact regrouping

$$E'_{l,h} = D_h \rho_l - (pI + P_l)E_{l,h} - E_{l,h}P_l - Q_{l,h}E_l - E_lQ_{l,h}. \quad (6.20)$$

Its initial value is zero: the past is fixed and $R_l(0,h) = D_h R_l(0,h) = 0$. Also $\|Q_{l,h}\| \leq C/T$. Apply Lemma 6.1 to (6.20). Its forcing has pointwise norm at most $C\eta_j/T$, by (6.17) and the first bound in (6.19). Its integrated norm is at most

$$C\eta_j + \frac{C}{T} \int_{t_j}^{t_j+T} \|E_l\| dt \leq C\eta_j.$$

Consequently

$$\sup_{t,h} \|E_{l,h}\| \leq \frac{C\eta_j}{T}, \quad \sup_h \int_{t_j}^{t_j+T} \|E_{l,h}\| dt \leq C\eta_j. \quad (6.21)$$

This argument supplies the derivative estimate directly, without an a priori estimate for $D_h P_l$.

To pass from coefficient errors to value transmissions, let $\mathcal{B}_l(t,h)$ be the pulse fundamental matrix up to time t , with value I at t_j , and set

$$F_l(t,h) = e^{-\int_{t_j}^t v_l(w) dw} \mathcal{B}_l(t,h), \quad G_l(t,h) = H_l((t-t_j)/T, h).$$

Then

$$F'_l = (R_l/T + E_l)F_l, \quad G'_l = (R_l/T)G_l, \quad F_l(t_j) = G_l(t_j) = I.$$

The integrated norms of R_l/T and $D_h R_l/T$ are uniformly bounded; those of E_l and $E_{l,h}$ are $O(\eta_j)$. The elementary integral inequality $u(t) \leq a + \int_{t_j}^t b(w)u(w) dw$, $b \geq 0$, implies $u(t) \leq a \exp(\int_{t_j}^t b)$: the right-hand side of the first inequality has derivative at most b times itself. Applying this inequality to the fundamental-matrix equations and their parameter derivatives bounds $F_l, G_l, D_h F_l, D_h G_l$ uniformly.

For completeness, $Z_l = F_l - G_l$ satisfies

$$Z'_l = (R_l/T + E_l)Z_l + E_l G_l, \quad Z_l(t_j) = 0.$$

The integral bound for E_l gives $\sup_t \|Z_l\| \leq C\eta_j$. Differentiating this equation introduces the additional terms

$$(D_h R_l/T + E_{l,h})Z_l + E_{l,h}G_l + E_l D_h G_l.$$

Their integrated norms are $O(\eta_j)$, using the preceding uniform bounds and (6.21). The same integral inequality gives $\sup_t \|D_h Z_l\| \leq C\eta_j$. Evaluation at the pulse end proves the first assertion of (6.14). At that end $R_l(1,h) = D_h R_l(1,h) = 0$. Combining $P_l = Q_l + E_l$, (6.15), (6.19), and (6.21) proves the second assertion. Every estimate used the earlier history only through the uniform reset and barriers of Lemma 6.2, proving the stated uniformity. $\square$

6.4 Removing the full diagonal leg

The pulse estimate describes values and radial derivatives at the pulse end. The radial derivative need not yet equal θ_l times the value, so diagonal angular coefficients on the next leg do not by themselves give a diagonal map on these data. We now factor the full leg exactly. This separates its possibly large unequal growth from the small error in the entrance derivative.

Let $\mathcal{T}_{l,j}(h)$ denote the value transmission of center-regular solutions from t_j to t_{j+1} . On the diagonal leg let $U_{l,j}$ and $V_{l,j}$ be the value propagation matrices from its start to its end for initial Cauchy data $(Y, Y') = (I, 0)$ and $(0, I)$, respectively. These matrices are diagonal in the fixed dwell basis. Define

$$D_{l,j} = U_{l,j} + \theta_l V_{l,j}, \quad K_{l,j} = D_{l,j}^{-1} V_{l,j}. \quad (6.22)$$

Thus $D_{l,j}$ is the value propagation on the entire leg for the reference data $(I, \theta_l I)$.

Lemma 6.4 (Exact normalization of the full leg and dwell growth). *The matrices $U_{l,j}$, $V_{l,j}$, and $D_{l,j}$ have positive diagonal entries, and*

$$0 \leq K_{l,j} \leq \theta_l^{-1} I. \quad (6.23)$$

They depend only on the prescribed diagonal leg, not on its current or earlier pulse parameters. The exact transmission identity is

$$D_{l,j}^{-1} \mathcal{T}_{l,j}(h) = [I + K_{l,j}(P_{l,e}(h) - \theta_l I)] \mathcal{B}_{l,j}(h). \quad (6.24)$$

Moreover, uniformly for the finite set of l, i ,

$$\log(D_{l,j})_{ii} = L_j d_{l,i} + O(1 + L_j j^{-2} + T_j) = L_j (d_{l,i} + o(1)). \quad (6.25)$$

Proof. Every operator on the leg is a function of I and $D_{j_0}^2$, and hence is diagonal in the fixed dwell eigenbasis. A scalar coordinate satisfies

$$y'' + p(t)y' - A_i(t)y = 0, \quad A_i(t) > 0.$$

Multiplying the equation for y' by the positive integrating factor $e^{\int p}$ gives

$$(e^{\int p} y')' = e^{\int p} A_i y.$$

Starting with $(y, y') = (1, 0)$, positivity of y makes y' nonnegative until any supposed first zero of y ; this prevents that zero. Starting with $(0, 1)$, the solution is positive for small positive time and the same argument keeps it positive. Thus the final entries of $U_{l,j}$ and $V_{l,j}$ are positive. Formula (6.22) gives positivity of $D_{l,j}$, and entry by entry

$$0 < \frac{(V_{l,j})_{ii}}{(U_{l,j})_{ii} + \theta_l (V_{l,j})_{ii}} \leq \theta_l^{-1}.$$

This proves (6.23).

For an incoming value vector y at the start of the period, the data at the pulse end are

$$(\mathcal{B}_{l,j} y, P_{l,e} \mathcal{B}_{l,j} y).$$

Their value at the end of the leg is

$$\mathcal{T}_{l,j} y = (U_{l,j} + V_{l,j} P_{l,e}) \mathcal{B}_{l,j} y.$$

Since $D_{l,j} = U_{l,j} + \theta_l V_{l,j}$, left multiplication by $D_{l,j}^{-1}$ gives (6.24). The order $K_{l,j}(P_{l,e} - \theta_l I)$ is part of this identity; no commutation with $P_{l,e}$ is used. The large matrix $D_{l,j}$ has disappeared from the error multiplier, leaving only the uniform bound (6.23).

To estimate $D_{l,j}$, take its i -th scalar solution with initial data $(1, \theta_l)$. The positivity just proved allows its logarithmic derivative $w = y'/y$ throughout the leg. It satisfies

$$w' = A_{l,i}(t) - p(t)w - w^2.$$

The scalar barrier argument in Lemma 6.2, with initial value θ_l , keeps w between fixed positive constants. During the dwell,

$$A_{l,i}(t) = \lambda_{l,i} + O(j^{-2}), \quad \lambda_{l,i} = d_{l,i}(d_{l,i} + p_\infty),$$

by (6.12) and the fixed value $q = q_*$. For $e = w - d_{l,i}$, subtraction gives

$$e' = F_{l,i}(t) - (p + w + d_{l,i})e, \quad F_{l,i} = A_{l,i} - \lambda_{l,i} - (p - p_\infty)d_{l,i} = O(j^{-2}).$$

The value of e at the dwell entrance is bounded, and its damping coefficient has a fixed positive lower bound. The scalar version of (6.6) yields

$$\int_{\text{dwell } j} |w - d_{l,i}| dt \leq C + CL_j j^{-2}.$$

The two ramps and the final rest have total duration $3T_j$, and their contribution to $\int w$ is $O(T_j)$. Since the solution starts at 1 and ends at $(D_{l,j})_{ii}$, integration of $w = (\log y)'$ gives the first equality in (6.25). The second follows from $T_j/L_j = j^{-1} \rightarrow 0$ and $L_j \rightarrow \infty$. All constants are uniform over the fixed finite list of coordinates. $\square$

6.5 Tuning every complete period exactly

The preceding identity puts the actual full-period map in the same local coordinate problem as the ideal word. Its remaining error tends to zero with one parameter derivative, uniformly over the past. This allows the target cycle to be attained successively in every period.

Proposition 6.5 (Exact full-period transmission). *For finite spectral data as specified at the end of Section 3, choose the word and compact pulse family above. There are a starting index j_0 and parameters $h_j \in K$, for every $j \geq j_0$, such that the center-regular value transmissions of the resulting metric satisfy*

$$\mathcal{T}_{l,j} = s_{l,j} D_{l,j} \Pi_l, \quad s_{l,j} > 0, \quad |\log s_{l,j}| \leq CT_j \quad (2 \leq l \leq L, j \geq j_0). \quad (6.26)$$

The positive diagonal matrices $D_{l,j}$ propagate the reference Cauchy data $(I, \theta_l I)$ over the entire leg following the pulse; they satisfy (6.25). The regular value equation has the bounds (6.9). All estimates used to choose j_0 are uniform over preceding admitted controls for this fixed finite family.

Proof. First fix an admitted past and vary the current parameter h . By (6.24),

$$e^{-\varphi_{l,j}} D_{l,j}^{-1} \mathcal{T}_{l,j}(h) = [I + K_{l,j}(P_{l,e}(h) - \theta_l I)] e^{-\varphi_{l,j}} \mathcal{B}_{l,j}(h).$$

The bound (6.23) and Lemma 6.3 imply

$$e^{-\varphi_{l,j}} D_{l,j}^{-1} \mathcal{T}_{l,j}(h) = H_l(1, h) + O_{C^1}(\eta_j). \quad (6.27)$$

The estimate is uniform in the preceding history. Each $\mathcal{T}_{l,j}$ is a value fundamental matrix of $Y' = P_l Y$, so it has positive determinant; so does $D_{l,j}$. Determinant normalization is therefore defined for their quotient. The matrices $H_l(1, h)$, $h \in K$, form a compact subset of the positive-determinant matrices. On a neighborhood of this compact set, $\mathcal{N}_l$ is smooth with bounded derivatives. Normalization is unchanged by the positive scalar $e^{-\varphi_{l,j}}$, so (6.27) gives

$$\Psi_j(h) := (\mathcal{N}_l(D_{l,j}^{-1} \mathcal{T}_{l,j}(h)))_{l=2}^L = \Phi(h) + O_{C^1}(\eta_j) \quad \text{in } \mathcal{G}. \quad (6.28)$$

The assertion is first an estimate for the ambient matrix entries and their derivatives, and hence also in any fixed smooth chart containing the compact image under consideration.

We spell out the uniform inversion. Use Euclidean norms on the parameter and coordinate spaces, and the associated operator norms for their derivatives. Choose a coordinate chart ζ near Π in $\mathcal{G}$ with $\zeta(\Pi) = 0$, and let $f = \zeta \circ \Phi$ and $A_0 = Df(0)$. By (6.4), A_0 is invertible. Choose a closed ball $\overline{B}_r$ centered at 0, contained in the interior of K , whose Φ -image lies compactly inside the chart and for which

$$\sup_{h \in \overline{B}_r} \|I - A_0^{-1} Df(h)\| \leq \frac{1}{4}.$$

For every sufficiently large j , uniformly in the admitted past, $f_j = \zeta \circ \Psi_j$ is defined on this ball and

$$\sup_{h \in \overline{B}_r} \|A_0^{-1}(Df_j(h) - Df(h))\| \leq \frac{1}{4}, \quad \|A_0^{-1} f_j(0)\| \leq \frac{r}{2}.$$

This follows from (6.28) and $\eta_j \rightarrow 0$. The map

$$\mathcal{C}_j(h) = h - A_0^{-1} f_j(h)$$

has derivative of norm at most $1/2$ on the ball. Also

$$\|\mathcal{C}_j(h)\| \leq \|\mathcal{C}_j(0)\| + \frac{1}{2}\|h\| \leq r \quad (h \in \overline{B}_r).$$

It is therefore a contraction of the closed ball into itself. Its iterates converge to a fixed point h_j , for which $f_j(h_j) = 0$, or $\Psi_j(h_j) = \Pi$.

All the preceding estimates were uniform over the past. Choose j_0 once, large enough for them and for the geometric input, after the fixed word and parameter ball have been chosen. The history before j_0 is round. Choose h_{j_0} by the contraction above, and repeat after each finite chosen history. This inductively defines an infinite admitted sequence and hence a single smooth metric of Proposition 5.1. Undoing determinant normalization at each fixed point gives

$$D_{l,j}^{-1} \mathcal{T}_{l,j} = s_{l,j} \Pi_l, \quad s_{l,j} = \det(D_{l,j}^{-1} \mathcal{T}_{l,j})^{1/N_l} > 0.$$

By (6.27), the determinant of $e^{-\varphi_{l,j}} D_{l,j}^{-1} \mathcal{T}_{l,j}$ stays bounded above and away from zero on the fixed compact family. Thus

$$\log s_{l,j} = \varphi_{l,j} + O(1) = O(T_j),$$

because v_l is uniformly bounded. This proves (6.26). The equality is for the complete period; no residual change of direction is passed to a later period. $\square$

7 Global growth and cone geometry

All parameters, the finite degree range, and the exact sequence of controls are now fixed. We first prove that the selected center-regular solutions satisfy the pointwise bound in Theorem 1.1. We then examine the cone limits, which explain why additional hypotheses in the comparison literature do not apply.

7.1 Every-point growth and the strict count

Proof of Theorem 1.1. We convert the exact cycles from Proposition 6.5 into polynomial growth. For each l with $M_l > 0$ and each $1 \leq i \leq M_l$, choose the center-regular solution with value vector $Y(t_{j_0}) = e_{l,i}$. Lemma 6.2 permits this choice. Near the origin it is r^l times an element of V_l , hence a homogeneous harmonic polynomial in Euclidean coordinates. It is smooth and harmonic there and, by (6.7), on the rest of the manifold.

Let σ_l be the cyclic permutation of $\{1, \dots, M_l\}$ underlying Π_l , and define $i_{j_0} = i$, $i_{j+1} = \sigma_l(i_j)$. Applying (6.26) sends e_{l,i_j} to a signed multiple of $e_{l,i_{j+1}}$, with absolute multiplier $s_{l,j}(D_{l,j})_{i_{j+1},i_j}$. Hence, with

$$\varepsilon_{l,i,j} = L_j^{-1} \log(D_{l,j})_{ii} - d_{l,i}, \quad \max_{l,i} |\varepsilon_{l,i,j}| \rightarrow 0$$

by (6.25), one has

$$\log \|Y(t_J)\| = \sum_{j=j_0}^{J-1} (\log s_{l,j} + j^7 (d_{l,i_{j+1}} + \varepsilon_{l,i_{j+1},j})). \quad (7.1)$$

Signs in Π_l do not affect this norm.

We verify the weighted average in (7.1). Group the sum of $j^7 d_{l,i_{j+1}}$ into complete cycles of length M_l . Within a cycle starting at j , $(j+r)^7 - j^7 = O(j^6)$ for $0 \leq r < M_l$, with a constant depending on this fixed length. Since the sum of $d_{l,i_{j+r+1}} - \bar{d}_l$ in a complete cycle is zero, that cycle contributes only $O(j^6)$ to the difference from the mean-weighted sum. Summing these errors and bounding the at most $M_l - 1$ final terms by $O(J^7)$ gives

$$\sum_{j=j_0}^{J-1} j^7 d_{l,i_{j+1}} = \bar{d}_l \sum_{j=j_0}^{J-1} j^7 + O(J^7).$$

Also $\sum_{j < J} |\log s_{l,j}| = O(\sum_{j < J} j^6) = O(J^7)$. The ε -terms sum to $o(J^8)$: for any positive number, choose a fixed late index after which their absolute values are below that number, and use $\sum_{j < J} j^7 = O(J^8)$. The finitely many earlier terms vanish after division by J^8 . Finally,

$$\sum_{j=j_0}^{J-1} j^7 = \frac{J^8}{8} + O(J^7), \quad t_J = \frac{J^8}{8} + O(J^7)$$

by (5.1). Therefore

$$\lim_{J \rightarrow \infty} \frac{\log \|Y(t_J)\|}{t_J} = \bar{d}_l < k. \quad (7.2)$$

The estimate extends to every radius. From $Y' = P_l Y$ and (6.9),

$$\|Y(t)\| \leq \exp(C_P(t - t_j)) \|Y(t_j)\| \quad (t_j \leq t \leq t_{j+1}).$$

The ratio $(t_{j+1} - t_j)/t_j$ tends to zero by (6.11). Thus (7.2) gives

$$\limsup_{t \rightarrow \infty} \frac{\log \|Y(t)\|}{t} \leq \bar{d}_l.$$

Choose α_l with $\bar{d}_l < \alpha_l < k$. For all sufficiently large $r = e^t$, $\|Y(\log r)\| \leq r^{\alpha_l}$. The fixed finite-dimensional space V_l has

$$\left| \sum_i Y_i e_{l,i}(\omega) \right| \leq C_l \|Y\| \quad \text{for every } \omega \in S^m,$$

for example by Cauchy–Schwarz and boundedness of the finitely many smooth basis functions. Since $d_g(0, x) = r(x)$, this gives $|u(x)| \leq C_u(1 + d_g(0, x))^k$ outside a compact set. Smoothness bounds u on that compact set and supplies the same inequality everywhere.

These functions are linearly independent: their restrictions to the sphere $t = t_{j_0}$ are the selected vectors of mutually orthogonal round spaces V_l . It follows that

$$\dim_{\mathbb{R}} \mathcal{H}_k(\mathbb{R}^{m+1}, g) \geq \sum_{l=2}^L M_l > \sum_{l=0}^k N_l.$$

The Euclidean identification was proved at the end of Section 3, so this is the required strict comparison.

Finally, the distance and volume statements in Proposition 5.1 give

$$\text{vol}_g B_g(0, R) = \text{vol}_{g_0}(S^m) \int_0^R r^m a(\log r)^m dr.$$

Because $a(t) \rightarrow a_*$, changing variables $r = Rz$ and using dominated convergence shows that this quantity divided by $\omega_{m+1} R^{m+1}$ tends to a_*^m ; here $\text{vol}_{g_0}(S^m) = (m+1)\omega_{m+1}$. This completes the proof. $\square$

7.2 Cone limits, conformal curvature, and cone degrees

Proposition 7.1 (Cone geometry and the union of degrees). *Let g be the metric constructed in the proof of Theorem 1.1, with link parameters a_*, q_* and comparison integer k . Every cone with link $G(a_*, q, J_0)$, $1 \leq q \leq q_*$, occurs as a pointed tangent cone at infinity. In particular the tangent cone is not unique. The metric g is not locally conformally flat. If $\mathcal{D}(M)$ denotes the union of nonnegative homogeneous harmonic degrees over all its tangent cones, then $\mathcal{D}(M)$ contains a half-line and a neighborhood of k .*

Proof. Centering a fixed logarithmic window in the round rests or nonround dwells and letting its center tend to infinity gives smooth annular limits with links $G(a_*, 1, J_0)$ and $G(a_*, q_*, J_0)$, respectively. They are nonisometric: the first is Einstein, whereas the second has distinct horizontal and vertical Ricci eigenvalues. Every intermediate ramp link also occurs as a limit, because the ramp durations diverge and their derivatives tend to zero. These annular limits extend to pointed cone limits including the vertex: the inner radial ball of radius δ has diameter at most 2δ in every rescaling and in the limit. Comparing paths on the complementary annuli, then letting $\delta \downarrow 0$, gives convergence of distances on each bounded ball. All these links have the same volume.

The additional local conformal flatness hypothesis in [2] also fails. Put $h_* = G(a_*, q_*, J_0)$. The nonround cone metric $dr^2 + r^2 h_*$ is conformal, under $t = \log r$, to the product $dt^2 + h_*$. For this product the Weyl tensor has mixed component

$$W_{0a0b} = -\frac{1}{m-1} \left(\text{Ric}_{h_*} - \frac{\text{Scal}_{h_*}}{m} h_* \right)_{ab}.$$

Indeed the product has zero mixed curvature and radial Ricci tensor; substitution in the trace decomposition of curvature gives this formula. The component is nonzero because the link has distinct horizontal and vertical Ricci eigenvalues. Vanishing of the Weyl tensor is conformally invariant and is preserved by smooth annular limits. If the constructed metric were locally conformally flat, its nonround cone limit would have zero Weyl tensor, contradicting the formula.

For maximal Hopf charge in degree l , its growth root $\alpha_l(q)$ is the nonnegative solution of

$$\alpha_l(q)(\alpha_l(q) + m - 1) = a_*^{-2} q^{1/m} [l(l + m - 1) + (q^{-1} - 1)l^2].$$

This root varies continuously with q . At $q = 1$ its asymptotic slope is a_*^{-1} ; at $q = q_*$ it is $a_*^{-1} q_*^{-(m-1)/(2m)}$, which is strictly smaller. More precisely, the endpoint roots are

$$\alpha_l(1) = a_*^{-1} l + O(1), \quad \alpha_l(q_*) = a_*^{-1} q_*^{-(m-1)/(2m)} l + O(1).$$

The difference of their slopes is positive, so for every sufficiently large l , $\alpha_{l+1}(q_*) < \alpha_l(1)$ and $\alpha_l(q_*) < \alpha_l(1)$. The continuous image of the q -interval contains the entire interval between these endpoints. These intervals overlap and have unbounded upper endpoints, so their union contains a half-line. In particular, equal cone-link volumes do not provide a discrete union spectrum.

The chosen integer k itself lies in this union. Indeed, (3.9) forces some selected degree $l > k$. Its smallest dwell root is below k , while its round root θ_l is greater than l because $a_* < 1$. Continuity supplies an intermediate cone root equal to k ; the strict endpoint inequalities in fact put a neighborhood of k in $\mathcal{D}(M)$. Xu's nonunique-cone three-circles theorem [16, Theorem 3.4] requires the comparison exponent to lie outside $\mathcal{D}(M)$, so it cannot be applied at k or at any sufficiently nearby exponent. The unique-cone spectral bounds in [8] also have a hypothesis that fails. A global logarithmic growth average, as in (7.2), need not equal the local homogeneous degree seen after normalization on each separate cone subsequence. $\square$

A An exact finite spectral selection

The analytic argument in Lemma 3.3 supplies all spectral data needed in the proof of Theorem 1.1. A supplementary integer calculation gives a concrete selection in ambient dimension 16, at the integer cutoff 50000. This calculation checks the finite spectral inequalities in (3.9); it does not construct the finite control word or the metric numerically. The analytic existence argument remains independent of it.

The parameters of the calculation are

$$\begin{aligned} m &= 15, & s &= 8, & k &= 50000, & q_* &= \frac{101}{100}, \\ c &= \frac{147}{1000}, & \alpha_*^2 &= \frac{99853}{100000}, & a_*^2 &= q_*^{1/15} \alpha_*^2. \end{aligned}$$

The endpoint curvature inequality is strict because

$$1 - \frac{2(q_* - 1)}{m - 1} - \alpha_*^2 = \frac{29}{700000} > 0.$$

The integer inequality $101 \cdot 99853^{15} < 100 \cdot 100000^{15}$ shows that $q_*(\alpha_*^2)^{15} < 1$, and hence $a_* < 1$. The decrease of the curvature threshold in the proof of Lemma 3.3 gives the margin for every $q \in [1, q_*]$. The radial scale also meets its required smallness condition:

$$-\log a_* \leq -\frac{1}{2} \log \alpha_*^2 < \frac{1 - \alpha_*^2}{2\alpha_*^2} < \frac{1}{1000}.$$

The cutoff integral in Proposition 5.1 is at least $\int_4^\infty (1+t)^{-5/4} dt = 4 \cdot 5^{-1/4} > 2$. It therefore gives $\mu < 1/2000$, in particular $b \geq -1/4$. These are parameter checks; the control and transmission arguments are still the analytic arguments of Sections 4–7.

Here is the arithmetic bound underlying the selection. Write $h = m-1$, $Q = 10^8$, and an angular eigenvalue as $\lambda = \text{num}/\text{den} > 0$, with integer numerator and denominator. If $\text{isqrt}(a) = \lfloor \sqrt{a} \rfloor$ for a nonnegative integer a , define

$$S = \text{isqrt} \left(\left\lfloor \frac{Q^2(h^2 \text{den} + 4 \text{num})}{\text{den}} \right\rfloor \right) + 1.$$

Then $S^2 \text{den} > Q^2(h^2 \text{den} + 4 \text{num})$, so the positive root of $d(d+h) = \lambda$ satisfies the strict rational bound

$$d < \frac{S - hQ}{2Q}.$$

For the charge indexed by b , the exact eigenvalue data are

$$\text{num} = 100000(101l(l+14) - (2b-l)^2), \quad \text{den} = 101 \cdot 99853.$$

The multiplicities are the integers in (3.2). The two equal contributions indexed by b and $l-b$ are combined, except for the middle term when they coincide. Increasing b from 0 to $\lfloor l/2 \rfloor$ puts the eigenvalues, and therefore the upper bounds for the exponents, in increasing order.

The supplied program `verification/counting-check.py` uses these integer multiplicities and upper bounds. It accepts an initial group only when its upper-bounded exponent sum is strictly less than k times its size. It may take part of a repeated eigenvalue, which corresponds to choosing that many vectors in its real eigenspace. Low degrees are selected in full whenever the exact inequality $l(l+14) < \alpha_*^2 k(k+14)$ certifies all their roots to be below k . In odd degree the zero Hopf charge is absent, so the full-degree test is a sufficient criterion rather than an exact characterization of the largest root. The individual search ends when the smallest root is at least k . Thus every counted group has a true mean strictly below k .

The resulting counts, using integer arithmetic, are

$$\begin{aligned} \text{selected dimension} &= 46785605044474340387088811257657817010714687366486217934111, \\ h_{50000}(\mathbb{R}^{16}) &= 46779709349146362239266126538609505511855492134986686636251, \\ \text{strict surplus} &= 5895695327978147822684719048311498859195231499531297860. \end{aligned}$$

The selected dimension omits degrees 0 and 1, as required in (3.9). Only the displayed strict integer comparison is used by the certificate; a decimal ratio is unnecessary.

These parameters and selections satisfy all the finite-data hypotheses at the end of Section 3. Applying Propositions 5.1 and 6.5, followed by the growth argument in Section 7, therefore gives the metric of Theorem 1.1 with $n = 16$ and $k = 50000$.

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